MOSFET

10.1 Introduction

The rapid strides of the semiconductor industry in recent years are due to its ability to incorporate more and more devices operating at higher and higher speeds in an Integrated Circuit (IC). Metal-Oxide-Semiconductor Field Effect Transistor (MOSFET) circuits occupy less silicon area and consume less power than their bipolar counterparts, making them the ideal choice as the device for Very Large Scale Integrated Circuits (VLSIs). In fact, the MOSFET is by far the most widely used semiconductor device today and is used for fabrication of Central Processing Unit (CPU) and memories in computers, Digital Signal Processors (DSPs) for Communication purposes and ICs for a variety of other applications.

In this chapter, we shall discuss the operating principles of the MOSFET and develop models for its characteristics. Fig. 10.1 shows the cross-sectional view of a MOSFET, which is a four-terminal device. The terminals are the Source (S), the Drain (D), the Gate (G) and the Bulk (B). As we can see, at the heart of the device is a Metal-Oxide-Semiconductor (MOS) structure, from which the MOSFET gets its name. For proper understanding of the device operation, we shall at first take up the two terminal MOS diode for discussion, followed by the MOSFET by including the additional Source and Drain terminals.

10.2 MOS diode

The MOS diode (Fig. 10.2) consists of an insulating layer (usually Silicon Dioxide) sandwiched between a metal and a semiconductor (usually Silicon). Interestingly, the structure consists of all the three different classes of solids, viz. metal,

semiconductor and insulator. We have already seen the nature of the energy band diagram of each of these materials (Section 1.2.2). However, it is now necessary to draw the energy band diagram of the composite MOS structure. For this purpose, the vacuum level is taken as the reference for drawing the relative positions of the energy levels of the different materials. We have already discussed in Section 1.3.10 that a metal is characterised by its **work function ($q\phi_m$ in eV)**, which is the energy required to raise an electron from its Fermi energy level to the vacuum level. For semiconductors and insulators, the **electron affinity ($q\chi$ in eV)**, which is the energy required to raise an electron from the bottom of the conduction band to the vacuum level, is a characteristic property of the material.

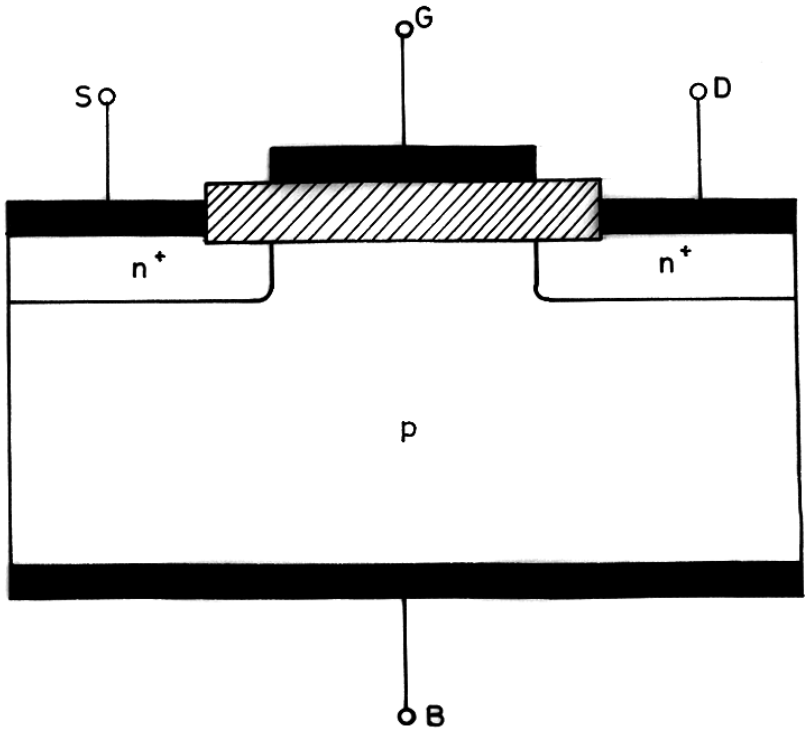


Fig.10.1 Cross-sectional view of a four-terminal MOSFET

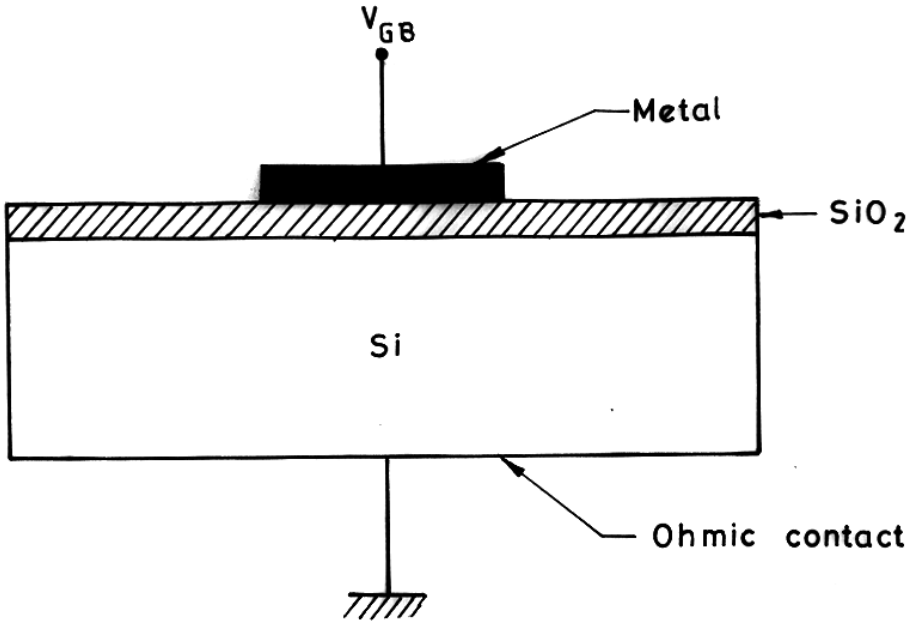


Fig. 10.2 Cross-sectional view of a two-terminal MOS diode

The work function of the semiconductor material (ϕ_s) is dependent on the doping concentration and can be calculated from the position of the Fermi level with respect to the bottom of the conduction band. Thus, for a p-type semiconductor, ϕ_s can be expressed as

$$\phi_s = \chi_s + (E_C - E_F)/q = \chi_s + E_g/2q + \phi_B \text{ ……..(10.1)}$$

where χ_s is the electron affinity and E_g is the bandgap of the semiconductor, q is the electronic charge (=$1.6\text{x}10^{-19}$ C) and ϕ_B is the energy difference between the Fermi level and the intrinsic level (E_i) given by

$$\phi_B = V_T \ln (N_A/n_i) \text{……..(10.2)}$$

where V_T is the thermal voltage (approximately 26mV at 300K), N_A is the acceptor impurity concentration in a p-type semiconductor substrate and n_i is the intrinsic carrier concentration ($1.5\text{x}10^{10}$ for silicon at 300K).

The work functions (ϕ) and electron affinities (χ) are expressed in units of Volts. The corresponding energies, $q\phi$ and $q\chi$, when expressed in eV, are numerically the same as already mentioned in Section 1.2.4. In other words, if ϕ = 3V, $q\phi = 1.6\text{x}10^{-19}$ x 3 J = 3eV. Similarly, if E_g = 1.1eV = 1.1 x $1.6\text{x}10^{-19}$ J, E_g/q = 1.1V.

Example 10.1. An ideal MOS diode (device MD1) is fabricated on a p-type silicon substrate having a doping concentration of $N_A=1\text{x}10^{16}/\text{cm}^3$. What is the work function of the silicon substrate (ϕ_s) at room temperature? (For silicon, E_g = 1.1eV, χ = 4.05 V, $n_i = 1.5\text{x}10^{10}/\text{cm}^3$ at 300K and V_T = 26mV at 300K)

From eqn. (10.2), $\phi_B = 0.026 \ln (1\text{x}10^{16}/1.5\text{x}10^{10}) = 0.35$ V.

From eqn. (10.1), $\phi_s = \chi_s + E_g/2q + \phi_B = 4.05 + (1.1/2) + 0.35 = 4.95$ V.

10.2.1 Operation of the ideal MOS diode

Let us at first consider an ideal metal-SiO_2-pSi system, which may be defined as a system which satisfies the following conditions:

(i) The metal work function (ϕ_m) is equal to the semiconductor work function (ϕ_s), or $\phi_{ms} = (\phi_m - \phi_s) = \phi_m - [\chi_s + E_g/2q + \phi_B] = 0$.

(ii) There are no charges in the oxide, or Q_{ox}=0. Hence, the only charges that exist in the device are those in the semiconductor and those with equal and opposite sign in the metal gate so that the device as a whole is neutral. In other words, $Q_G + Q_S = 0$. Here, Q_G, Q_{ox} and Q_S are the charge densities per unit area in the gate, oxide and semiconductor respectively.

(iii) There is no carrier transport through the oxide, as its resistivity is infinitely large.

Ideal MOS diode at zero bias

Fig. 10.3(a) shows the band diagram of the ideal MOS structure at zero gate-to-bulk bias, i.e. $V_{GB} = V_G - V_B = 0$. The vacuum level is flat, since there is no voltage dropped in the device. Since $\phi_{ms} = \phi_m - \phi_s = 0$, the Fermi energy levels in the metal (gate) and semiconductor coincide. The notations in the band diagram in Fig. 10.3(a) can be modified and represented as in Fig. 10.3(b). In Fig. 10.3(b), $q\phi_m'$ is the **metal-oxide barrier energy**, which is the energy required to move an electron from the metal Fermi level to the conduction band of the oxide, where

$$\phi_m' = \phi_m - \chi_{SiO2} \quad \text{.....(10.3)}$$

Similarly, the **silicon-oxide barrier energy** is the energy required to move an electron from the silicon conduction band to the conduction band of the oxide and is given by $q\chi_S'$ where

$$\chi_S' = \chi_S - \chi_{SiO2} \quad \text{.....(10.4)}$$

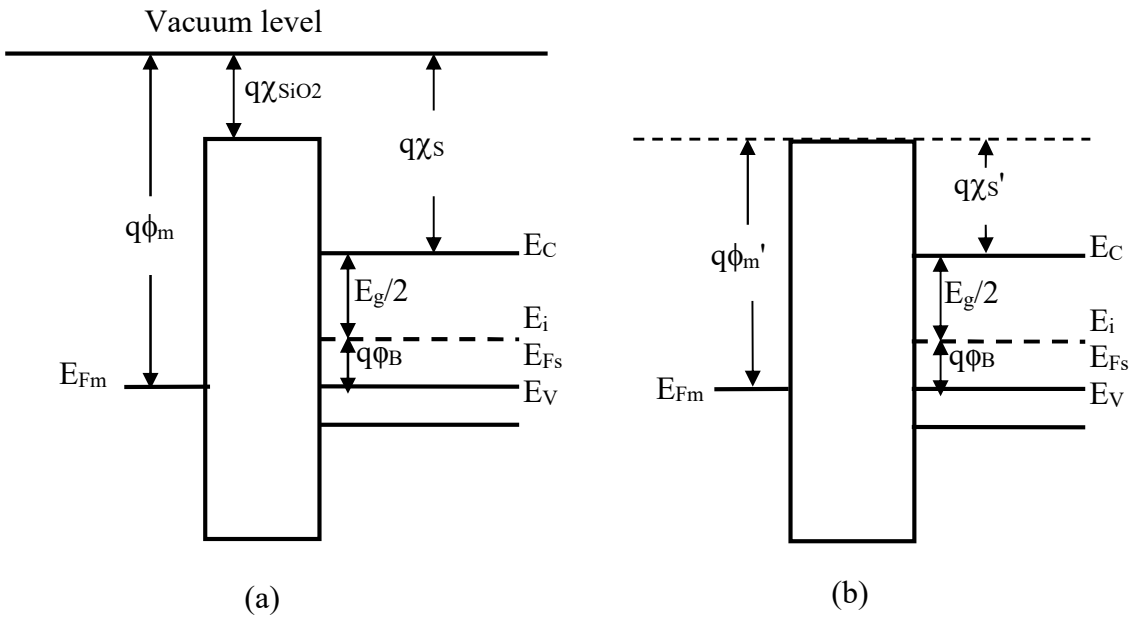


Fig. 10.3 Band diagram of an ideal MOS diode at $V_{GB} = 0V$ drawn using (a) ϕ_m, χ_{SiO2}, χ_S and (b) ϕ_m' ,χ_S'.

It must be remembered that ϕ_m' and χ_S' are constants for a particular material system and does not change with application of bias. As it is more convenient, we shall use the constants ϕ_m' and χ_S' to draw the band diagrams in this chapter.

Ideal MOS diode at $V_{GB} > 0V$

Fig.10.4 shows the band diagram of an ideal MOS diode with applied bias $V_{GB} > 0V$. The applied voltage V_{GB} is the sum of two parts: ψ_{ox} dropped in the oxide and ψ_s dropped in silicon. This is reflected in the figure, with a band bending ψ_{ox} in the oxide and ψ_s in silicon. A point to be noted in the figure, is that the metal Fermi level (E_{Fm}) is at a lower energy level compared to the Fermi level in the semiconductor (E_{Fs}) by qV_{GB}, since the metal is at a higher potential with respect to the semiconductor by V_{GB} volts. (Remember: The energy band diagram reflects the electronic potential, which is the negative of electrostatic potential.) We shall show that this is indeed true. Adding up the energies in the band diagram, we have

$$E_{Fm} + q\phi_m' + q\psi_{ox} = E_{Fs} + q\phi_B + E_g/2 - q\psi_s + q\chi_S' = E_{Fs} - q\psi_s + q\phi_s'$$

Now, for the ideal MOS diode, $q\phi_m' = q\phi_s'$ and since $V_{GB} = \psi_{ox} + \psi_s$, we have

$$E_{Fs} - E_{Fm} = qV_{GB} \quad \text{.....(10.5)}$$

So, for drawing energy band diagram of the MOS structure, the difference in the energy levels of the semiconductor and metal Fermi levels must be made equal to the applied bias V_{GB}.

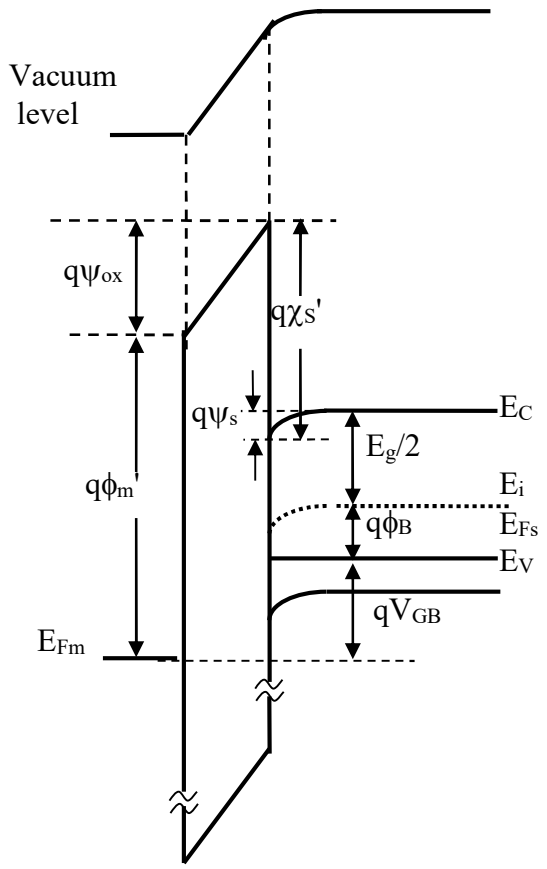


Fig. 10.4 Band diagram of an ideal MOS diode at $V_{GB} > 0V$

Another point to be noted is that the Fermi level has been drawn flat in the semiconductor. This is because there can be no current flowing through the oxide and hence no steady state current in the semiconductor substrate. As we have already seen in Section 1.3.9, a flat Fermi level implies zero current. The vacuum level is also drawn in Fig.10.4. The vacuum level shows the variation of the electronic potential in the device. Since the electric field, which is the derivative of the electronic potential, is a finite quantity, we find that the vacuum level is indeed continuous, although there may be discontinuities in the band edges at the interface of two dissimilar materials.

We also see in the band diagram, that due to the potential drop of ψ_s in the semiconductor, the intrinsic Fermi level (E_i) is bent towards the Fermi level (E_F) at the semiconductor surface, or in other words, the energy difference ($E_i - E_F$) is reduced. At the semiconductor surface, $(E_i - E_F)_s = q(\phi_B - \psi_s)$. The hole concentration at the semiconductor surface (p_{ps}) is therefore given by

$$p_{ps} = n_i \exp\{(\phi_B - \psi_s)/V_T) = p_{po} \exp\{(-\psi_s)/V_T) \quad \ldots\ldots\ldots(10.6)$$

where p_{po} is the equilibrium hole concentration in the p-type bulk silicon given by $p_{po} \approx N_A = n_i \exp(\phi_B/V_t)$. On the other hand, the electron concentration at the semiconductor surface (n_{ps}) is given by

$$n_{ps} = n_i \exp\{(\psi_s - \phi_B)/V_T) = n_{po} \exp\{(\psi_s)/V_T) \quad \ldots\ldots\ldots\ldots(10.7)$$

where n_{po} is the equilibrium electron concentration in bulk silicon given by $n_{po} = (n_i^2/p_{po}) = n_i \exp(-\phi_B/V_T)$. So we find that with the application of a positive gate voltage, the majority carrier concentration is reduced, while the minority carrier concentration increases at the semiconductor surface. It may also be noted that due to the variation of $(E_i - E_F)$ near the semiconductor surface, the hole (electron) concentration is minimum (maximum) at the surface and gradually increases (reduces) towards the thermal equilibrium value in the bulk of the semiconductor.

Example 10.2. In the device MD1, calculate the hole concentration (p_{ps}), electron concentration (n_{ps}) and the charge density (ρ) at the semiconductor surface for (i) ψ_s = 100mV and (ii) ψ_s = 200mV.

(i) For ψ_s = 100 mV

From eqn. (10.6), $p_{ps} = p_{po} \exp\{(-\psi_s)/V_T) = 10^{16} \exp(-100/26) = 2.1\text{x}10^{14}/\text{cm}^3$

From eqn. (10.7), $n_{ps} = (n_i^2/p_{po})\exp\{(\psi_s)/V_T) = (1.5x10^{10})^2/10^{16}\ \exp(100/26) = 1x10^6/cm^3$

Now, $\rho = q(N_D^+ - N_A^- + p - n) = 1.6x10^{-19}$ x $(-1x\ 10^{16} + 2.1x10^{14} - 1x10^6)$

$= 1.6x10^{-19}$ x $(-0.979 \times 10^{16}) = -1.566x10^{-3} C/cm^3 \approx -qN_A$

(i) For $\psi_s = 200$ mV

From eqn. (10.6), $p_{ps} = p_{po} \exp\ \{(-\psi_s)/V_T) = 10^{16} \exp\ (-200/26) = 4.56x10^{12}/cm^3$

From eqn. (10.7), $n_{ps} = (n_i^2/p_{po})\ \exp(\psi_s)/V_T) = (1.5x10^{10})^2/10^{16}\exp(200/26) = 4.93x10^7/cm^3$

Now, $\rho = q(N_D^+ - N_A^- + p - n) = 1.6x10^{-19}$ x $(-1x\ 10^{16} + 4.56x10^{12} - 4.93x10^7)$

$= 1.6x10^{-19}$ x $(-0.9995 \times 10^{16}) = -1.599x10^{-3} C/cm^3 \approx -qN_A$

So we see from Example 10.2 that even when ψ_s is as small as 100mV, $\rho \approx -qN_A$ at the semiconductor surface, since the electron, hole and donor impurity concentrations are very much less than the acceptor impurity concentration. With increasing ψ_s, ρ further approaches $-qN_A$. Therefore, we can say that due to the application of a positive voltage at the gate of the MOS capacitor, negative charges are induced in the semiconductor as the holes are repelled from the surface and the acceptor ions are uncovered. A depletion (space charge) layer is created at the semiconductor surface where the charge density is given by

$$\rho = -q N_A \quad \text{.....(10.8)}$$

The neglect of charges due to the carriers, as in eqn.(10.8), is often referred to as the **depletion approximation**. It may be mentioned here that this approximation is strictly not true at the edge of the depletion region where the hole concentration is comparable to the acceptor ion concentration. But in order to calculate the depletion width, for the sake of simplicity, we shall assume that eqn.(10.8) holds in the entire depletion layer. We have

already derived a relation for the potential drop across a depletion layer of thickness W having a constant charge concentration [refer to eqn.(4.18)]. Using the same relation, the depletion layer width W when the drop in the semiconductor is ψ_s can be expressed as

$$W = \sqrt{\frac{2\varepsilon_s \psi_s}{qN_A}} \quad \ldots\ldots\ldots(10.9)$$

So as the semiconductor surface potential increases, the depletion layer width W increases. From the depletion approximation, the total charge density per unit area in the semiconductor (Q_S) is approximately equal to the number of acceptor ions per unit area in the depletion layer. This is often referred to as the **bulk charge density (Q_B)**, and is given by

$$Q_S \approx Q_B = -qN_A W = -qN_A \sqrt{\frac{2\varepsilon_s \psi_s}{qN_A}} = -\sqrt{2qN_A \varepsilon_s \psi_s} \quad \ldots\ldots\ldots..(10.10)$$

From the above equation we see that the semiconductor charge also increases with ψ_s.

We shall now evaluate the potential drop across the oxide ψ_{ox} corresponding to a drop of ψ_s in the semiconductor. For this, we shall use Gauss Law, which is written as

$$\oint_s D.ds = \int_{vol} \rho.dv \quad \ldots\ldots\ldots(10.11)$$

where the electric displacement vector D is related to the electric field ε through the relation D = εε under static conditions. When expressed in words, Gauss Law states that the total normal electric flux coming out of a closed surface equals the charge enclosed by the surface. Let us consider Fig. 10.5, which shows an imaginary Gaussian surface enclosing part of the MOS capacitor. The electrical field lines originate from the positive charges in the gate and terminate on the negative charges in the semiconductor. Deep inside the semiconductor, the electric field goes to zero. Hence, there are no flux lines

coming out of the bottom of the Gaussian surface. Also, there are no flux lines coming out of the sides, since the electric field lines run parallel to them. The only flux lines intersecting the Gaussian surface are at the top. If E_{ox} is the electric field in the oxide (it is a constant since there is no charge in the oxide), and assuming unit surface area, from eqn. (10.11) we have

$$\varepsilon_{ox}\, \varepsilon_{ox} = -Q_S \quad \text{.....(10.12)}$$

If the thickness of the oxide is t_{ox}, then for a drop of ψ_{ox} across it, we have $\varepsilon_{ox} = \psi_{ox}/ t_{ox}$. Hence, from eqn. (10.12), we have

$$\psi_{ox} = -Q_S \frac{t_{ox}}{\varepsilon_{ox}} = \frac{-Q_S}{C_{ox}} \quad \text{.....(10.13)}$$

where **$C_{ox} = \varepsilon_{ox}/ t_{ox}$ is the oxide capacitance per unit area**. Since the applied voltage V_{GB} is the sum of the potentials dropped in the oxide and semiconductor, we may write using eqns. (10.10) and (10.13)

$$V_{GB} = \psi_s + \psi_{ox} = \psi_S + \frac{qN_A W}{C_{ox}} = \psi_S + \frac{\sqrt{2qN_A \varepsilon_s \psi_S}}{C_{ox}} \quad \text{.....(10.14)}$$

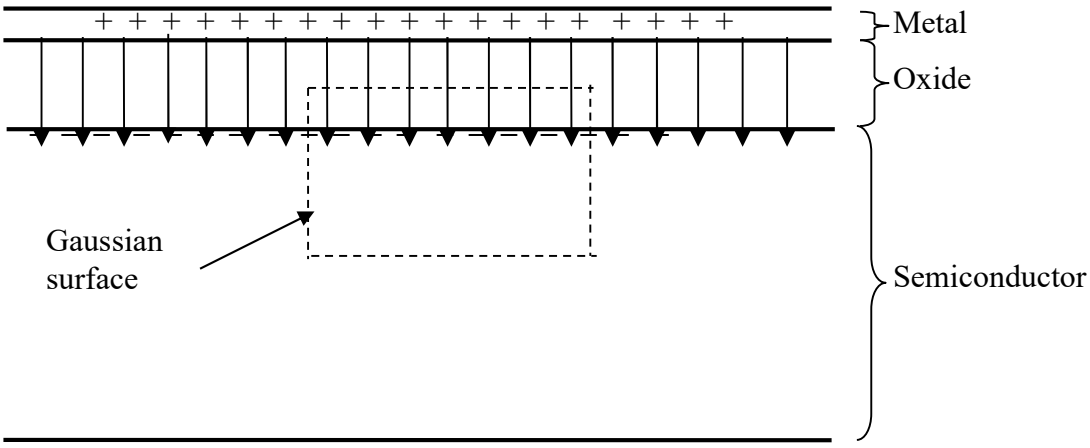


Fig. 10.5 A MOS structure showing the charges and electric field lines at $V_{GB} > 0V$. The electric field lines cut the imaginary Gaussian surface only at the top.

From eqn. (10.14), we see that as ψ_s increases, V_{GB} increases. Conversely, as V_{GB} increases, both ψ_{ox} and ψ_s increases. As ψ_s increases, the band is bent further, and when $\psi_s = \phi_B$, the Fermi level at the surface is at the intrinsic Fermi level. At this point, $p_{ps} = n_{ps} = n_i$. As ψ_s increases above ϕ_B, the electron concentration increases to above n_i and the hole concentration reduces to below n_i. Thus the electrons become the majority carriers and the semiconductor surface is said to be inverted. But even then both electron and hole concentrations are much less than the acceptor impurity concentration and the depletion layer charge is given by eqn. (10.8), i.e. $\rho = -q\, N_A$. Also, the depletion layer goes on widening with increasing ψ_s. When $\psi_s = 2\phi_B$, from eqns. (10.2), (10.6) and (10.7) we have $p_{ps} = n_i^2/N_A$ and $n_{ps} = N_A$. That is, when $\psi_s = 2\phi_B$, the electron concentration at the surface becomes equal to the hole concentration in the bulk of the p-type semiconductor substrate. We call this point the onset of strong inversion. Fig.10.6 shows

the band diagram and the charge distribution in the semiconductor at the onset of strong inversion. We see that the semiconductor charge (Q_s) is actually the sum of two components: the bulk charge ($Q_B = -qN_AW$), which is due to uncovering of the acceptor impurities in the depletion region and the electron charge ($Q_n = -q\int_0^W n(x)dx$), which is due to the electron concentration at the semiconductor surface. For $\psi_s \leq 2\phi_B$, we have assumed that $Q_s \approx Q_B$. This is because, in this range $Q_n << Q_B$. Even when $\psi_s = 2\phi_B$, although $n_{ps} = N_A$, for most of the depletion region $n(x) << N_A$, and so the assumption $Q_s \approx Q_B$ is valid. But when $\psi_s > 2\phi_B$, Q_n becomes comparable to Q_B. We have already seen that Q_B is proportional to $\sqrt{\psi_s}$ while the electron concentration at the semiconductor surface is proportional to exp (ψ_s/V_T). So a small change in ψ_s results in a large increase in n_{ps}. In fact, at room temperature, n_{ps} increases by an order of magnitude for every 60mV increase in ψ_s, since exp(60/26) ≈ 10. A large increase in n_{ps} results in a correspondingly large increase in Q_s and ψ_{ox} as can be seen from eqn. (10.13). So, in strong inversion a small increase in ψ_s results in a large increase in ψ_{ox}. Since $V_{GB} = \psi_s + \psi_{ox}$, once strong inversion is reached, further increase in V_{GB} is absorbed mostly by ψ_{ox} while ψ_s remains almost equal to $2\phi_B$. The depletion layer width therefore saturates at a maximum value of W_{max}. Substituting $\psi_s = 2\phi_B$ in eqn.(10.9) we have

$$W_{max} = \sqrt{\frac{2\varepsilon_s(2\phi_B)}{qN_A}} \quad \text{.....(10.15)}$$

Physically, the sheet of electrons in the inversion layer at the semiconductor surface can be thought of as screening the semiconductor interior from changes in gate potential.

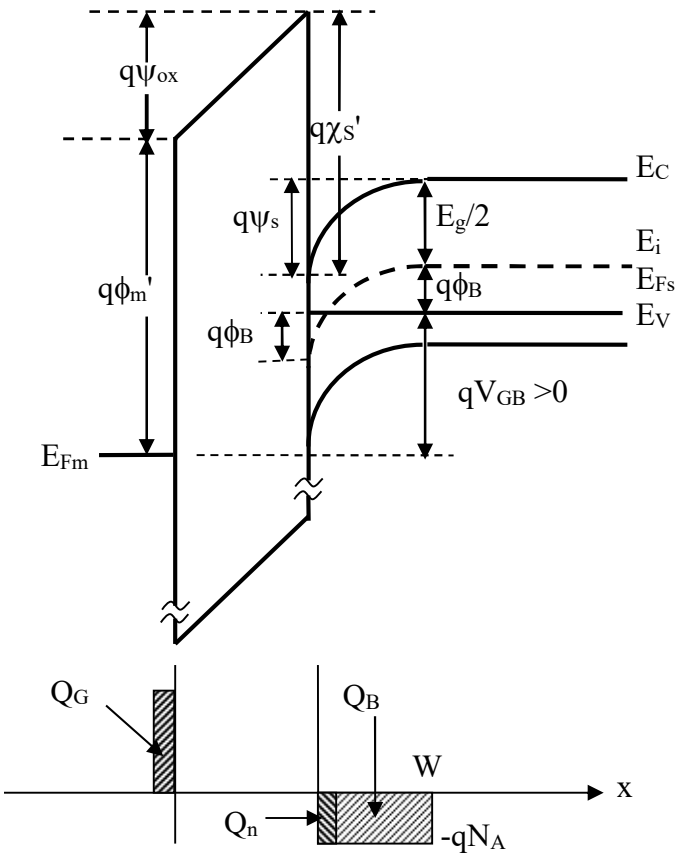


Fig. 10.6 Band diagram and charge distribution in the semiconductor at the onset of strong inversion

The value of V_{GB} that must be applied to just create a condition of strong inversion is known as the threshold voltage (V_{th}) of the MOS diode, which is one of the most important parameters of any MOS device. The value of V_{th} of the ideal MOS diode can now be obtained by substituting $\psi_s = 2\phi_B$ in eqn. (10.14), so that

$$(V_{th})_{ideal} = 2\phi_B + \frac{\sqrt{2qN_A\varepsilon_s(2\phi_B)}}{C_{ox}} \quad \text{.....(10.16)}$$

Example 10.3. In the device MD1, if the gate oxide thickness (t_{ox}) is 1000Å, calculate (i) the maximum depletion width (W_{max}) and (ii) the threshold voltage (V_{th}). [ε_r = 11.9 for Si and ε_r = 3.9 for SiO_2]

For device MD1, ϕ_B = 0.35V(from Ex. 10.1).

(i) From eqn. (10.15), $W_{max} = \sqrt{\frac{2\varepsilon_{Si}(2\phi_B)}{qN_A}} = \sqrt{\frac{2x11.9x8.85x10^{-14}x0.7}{1.6x10^{-19}x10^{16}}} = 0.304\mu m$

(ii) $C_{ox} = \varepsilon_{ox}/ t_{ox} = (3.9x8.85x10^{-14})/(1000x10^{-8}) = 3.45x10^{-8}$ F/cm^2.

From eqn.(10.16),

$$(V_{th})_{ideal} = 2 \times 0.35 + \frac{\sqrt{2x1.6x10^{-19}x10^{16}x11.9x8.85x10^{-14}x0.7}}{3.45x10^{-8}} = 2.11V$$

Alternately, at $V_{GB} = (V_{th})_{ideal}$, $\psi_{ox} = \frac{qN_A W_{max}}{C_{ox}}$

$$= \frac{1.6x10^{-19}x10^{16}x0.304x10^{-4}}{3.45x10^{-8}} = 1.41V$$

Therefore, $(V_{th})_{ideal} = \psi_{ox} + \psi_s = 1.41 + 0.7 = 2.11V$

Q. 10.1. How does V_{th} depend on N_A and t_{ox}?

V_{th} increases with increase in N_A since both ψ_s (=$2\phi_B$) and ψ_{ox} ($= \frac{\sqrt{2qN_A\varepsilon_s(2\phi_B)}}{C_{ox}}$) at threshold increases with increase in N_A. With increase in t_{ox}, C_{ox} decreases and consequently ψ_{ox} increases. Hence, V_{th} also increases with increase in t_{ox}.

Ideal MOS diode at $V_{GB} < 0V$

Fig.10.7 shows the band diagram of an ideal MOS diode with applied bias $V_{GB} < 0$. It may be noted in the figure that E_{Fm} has moved up by qV_{GB} with respect to E_{Fs}, since the metal is now at a lower potential with respect to the semiconductor by V_{GB} volts. Also, the figure shows that V_{GB} is the sum of two parts: ψ_{ox} dropped in the oxide and ψ_s dropped in silicon. In this case, the potential drop ψ_s has resulted in the valence band edge coming closer to the Fermi level, which means that the hole concentration at the semiconductor surface has increased. This is known as the accumulation condition.

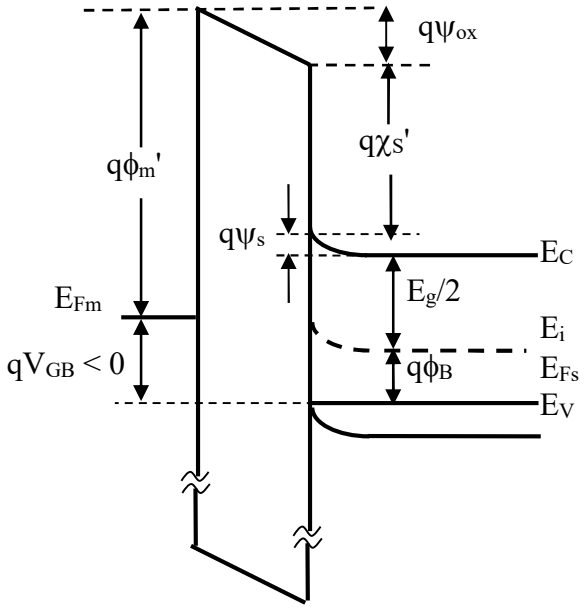


Fig. 10.7 Band diagram of an ideal MOS diode at $V_{GB} < 0V$

In summary, when $V_{GB} = 0$, $\psi_s = 0$ and the bands are flat giving rise to the **flat band** condition. Application of positive voltage on metal induces negative charges in the semiconductor. Initially this is due to holes being repelled from the semiconductor surface. On further increase of V_{GB}, the negative charges are due to the minority carrier electrons, which are attracted to the semiconductor surface and remain there, as they

cannot cross the insulator. When $0 < \psi_s \leq 2\phi_B$, the semiconductor surface is depleted of carriers giving rise to the **depletion** condition. The region of operation $\phi_B < \psi_s \leq 2\phi_B$ is also referred to as the **weak inversion** condition, since the electron concentration at the semiconductor surface is now more than the concentration of holes, which are the majority carriers in the silicon bulk. **Strong inversion** condition is said to exist when $\psi_s \geq 2\phi_B$. On the other hand, application of negative voltage at the gate induces positive charges at the semiconductor surface by attracting holes (majority carriers) to the semiconductor surface. This region of operation, when $\psi_s<0$ is known as the **accumulation** condition, since the majority carriers accumulate at the semiconductor surface.

In this section, we have only considered MOS diodes fabricated on p-type semiconductor substrates. For MOS diodes fabricated on n-type semiconductor substrates, the operation is identical except that the polarity of voltages must be interchanged. For example, a positive applied voltage results in accumulation condition and the threshold voltage is negative in these devices.

10.2.2 Operation of MOS diode with $\phi_{ms}\neq0$, $Q_{ox}=0$

Fig. 10.8(a) shows the energy band diagram of a MOS diode with $\phi_{ms} \neq 0$ and $Q_{ox} = 0$ biased at $V_{GB} = 0V$. When the gate and the semiconductor are short-circuited, there is an initial flow of charge till the Fermi energy levels are aligned, i.e. $E_{Fs} - E_{Fm} = 0$. As we see in Fig. 10.8(a), for devices with $\phi_{ms} = \phi_m - \phi_s < 0$, the semiconductor surface becomes negatively charged. So, in such a device, the semiconductor surface may be inverted even when $V_{GB} = 0V$. On the other hand, if $\phi_{ms} > 0$, there would be accumulation of holes at the semiconductor surface when $V_{GB}=0$. We can see from Fig. 10.8(a) that

$$q\phi_m' + q\psi_{ox} = q\phi_B + E_g/2 - q\psi_s + q\chi_s' = q\phi_s' - q\psi_s$$

$$\text{or, } \phi_{ms} = \phi_m' - \phi_s' = -(\psi_{ox} + \psi_s) \ldots\ldots\ldots\ldots (10.17)$$

So, the sum of the potential drops across the oxide and semiconductor is equal to the difference between the work functions of the semiconductor and metal. The fact that there is a potential drop in the semiconductor and oxide when $V_{GB} = 0$ is similar to the presence of the built-in potential in a p-n junction, which is also equal to the work function difference between two regions of the device.

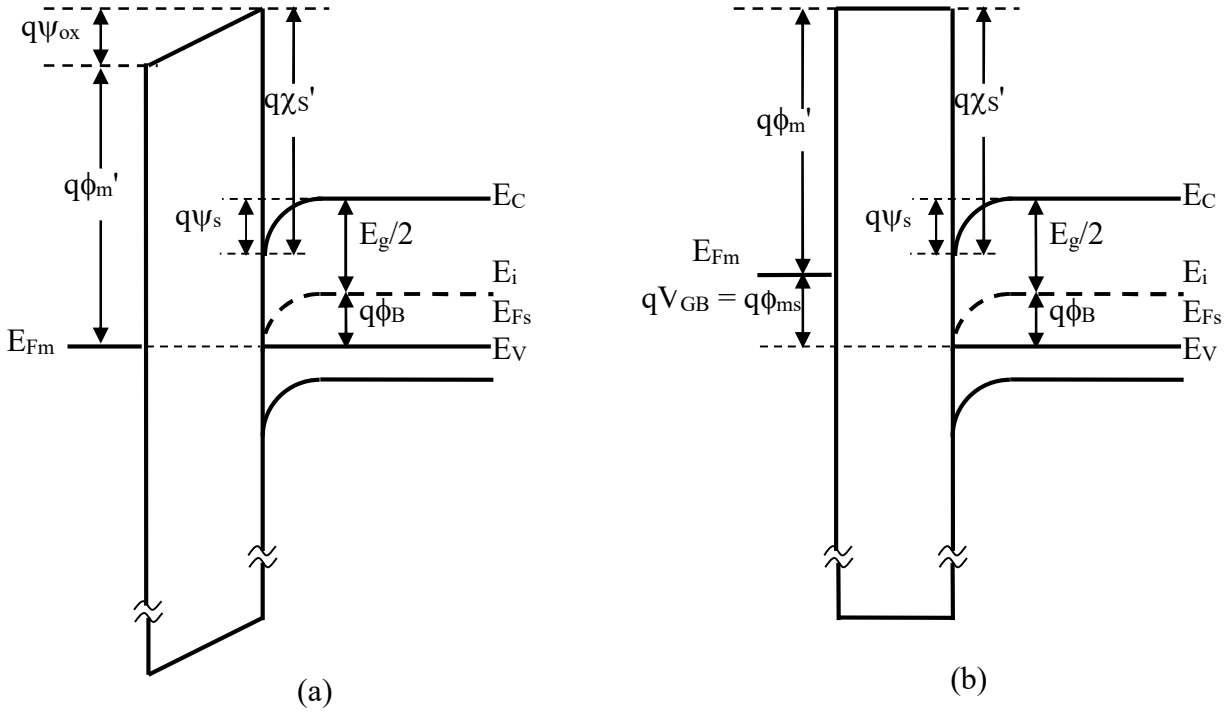


Fig. 10.8 Energy band diagram of a MOS diode with $\phi_{ms} \neq 0$ and $Q_{ox}=0$biased at (a) $V_{GB}=0V$ and (b) $V_{GB} = V_{FB}$.

In modern MOS technology, n^+ doped polysilicon is often used as the gate material. In this case, the electron affinities of both the gate material and the semiconductor are identical. So the work function difference ϕ_{ms} is actually the difference of the Fermi energy level with respect to the conduction band edge in the n^+ polysilicon

gate and the silicon substrate. In n^+ polysilicon, we may assume that the Fermi level coincides with the conduction band edge. So, for an n^+ polysilicon gate MOS diode fabricated on a p-type silicon substrate, we have

$$\phi_{ms} = -E_g/2q - \phi_B \quad \ldots\ldots\ldots\ldots(10.18)$$

If $\phi_{ms} < 0$, it is possible to revert back to the flat-band condition, i.e. $\psi_s = 0$, by applying $V_{GB} < 0V$. By drawing the energy band diagram using eqn.(10.5) as shown in Fig. 10.8(b), we find that the flat-band condition is achieved if $V_{GB} = \phi_{ms}$. Thus, when $\phi_{ms} \neq 0$,

$$V_{FB} = \phi_{ms} \quad \ldots\ldots\ldots\ldots(10.19)$$

where V_{FB} is the gate voltage with respect to bulk required to achieve flat-band condition and is called the flat-band voltage. As shown in Fig. 10.8(b), when $V_{GB} = V_{FB}$, $\psi_{ox} = \psi_s = 0$. Now, if $V_{GB} < V_{FB}$, $\psi_s < 0$ giving rise to the accumulation condition. Also, to have $\psi_s > 0$, i.e. for the depletion or inversion conditions, V_{GB} must be greater than V_{FB}. So, to get the identical condition in the semiconductor as in the ideal diode an extra voltage of V_{FB} must be applied. In other words, wherever we have V_{GB} in the analysis of ideal diode, we may use $V_{GB} - V_{FB}$, or we may write

$$V_{GB} - V_{FB} = \psi_s - \frac{Q_S}{C_{ox}} \quad \ldots\ldots\ldots\ldots\ldots(10.20)$$

Using the condition that $\psi_s = 2\phi_B$ when $V_{GB} = V_{th}$, we have from eqn.(10.10) and (10.20)

$$V_{th} = V_{FB} + 2\phi_B + \frac{\sqrt{2qN_A\varepsilon_s(2\phi_B)}}{C_{ox}} = V_{FB} + (V_{th})_{ideal} \quad \ldots\ldots(10.21)$$

Q10.2 In Fig.10.8(a), we see that when $V_{GB} = 0V$, there is potential drop across the oxide as well as the semiconductor, i.e . $\psi_{ox} \neq 0$ **and** $\psi_s \neq 0$. Is it not possible to take care of the work function difference by ψ_{ox} alone, so that $\psi_s = 0$, so that flat band condition exists.

No, it is not possible to take care of the work function difference by ψ_{ox} alone. From Gauss Law (eqn. (10.11)), we know that in the absence of any charge in the oxide, any electric field in the oxide will result in a corresponding electric field in the semiconductor. Thus if $\varepsilon_{ox} \neq 0$, an electric field in the semiconductor must exist, so that $\psi_s \neq 0$.

Example 10.4. The MOS diodes MD2 and MD3 are identical to the ideal MOS diode MD1, except that MD2 uses aluminium as gate metal and MD3 uses n^+ polysilicon. Calculate the threshold voltages of MD2 and MD3. [For Al, work function(ϕ) = 4.1V]

For device MD1, $\phi_B = 0.35V$ and $\phi_s = 4.95V$(Ex. 10.1) and $(V_{th})_{ideal} = 2.11V$(Ex. 10.3)

(i) For MD2, $V_{FB} = \phi_{ms} = (\phi_m - \phi_s) = 4.1 - 4.95 = -0.85V$

From eqn. (10.21), $V_{th} = V_{FB} + (V_{th})_{ideal} = -0.85 + 2.11 = 1.26V$.

(ii) For MD3, from eqn.(10.18), $\phi_{ms} = -E_g/2q - \phi_B = -0.55 - 0.35 = -0.9V$.

Therefore $V_{th} = V_{FB} + (V_{th})_{ideal} = -0.9 + 2.11 = 1.21V$.

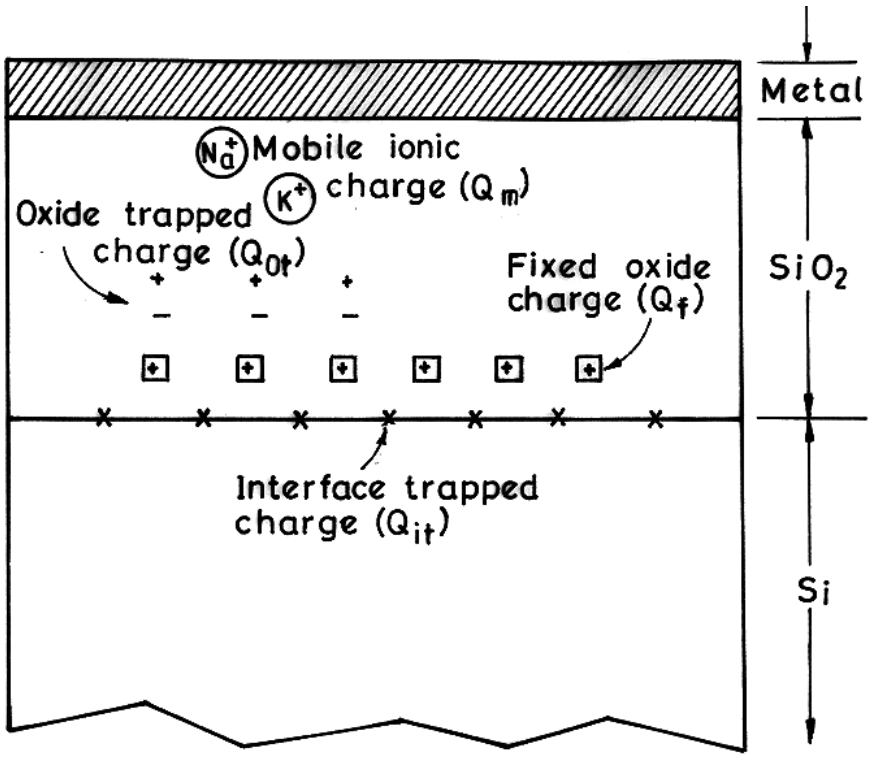


Fig.10.9 The different types of traps and charges and their location[1]

10.2.3 Operation of MOS diode with $\phi_{ms}\neq 0$, $Q_{ox}\neq 0$

In addition to the metal-semiconductor work function difference, another non-ideality is the presence of traps at the semiconductor-oxide interface and charges in the oxide. The different types of traps and charges and their location are shown in Fig. 10.9.

A silicon surface is a discontinuity in an otherwise periodic crystal lattice. From quantum mechanics, it can be shown that the discontinuity at a clean semiconductor surface gives rise to a number of allowed states in the forbidden gap. These surface states are due to the unsatisfied covalent bonds at the surface (also known as dangling bonds). The density of the surface states is equal to the density of the dangling bonds, i.e. about

$10^{15}/cm^2$. On oxidation of the surface layer, most of the surface atoms are bound to oxygen resulting in a sharp reduction of the density of states at the interface of silicon and its oxide to about $10^{11} – 10^{12}/cm^2$. In a silicon surface of <100> orientation, the interface state density is about an order of magnitude smaller than in <111>. Thus all MOS devices are fabricated on silicon substrates with <100> orientation. A low temperature (450ºC) hydrogen anneal is often used to reduce the density of states further by the formation of Si-H bonds at the interface. These **interface states** reside within 25Å of the Si-SiO_2 interface, and play an important role in determining device behaviour. The interface states are distributed nonuniformly throughout the forbidden gap. Therefore they are quantified as interface trap density (D_{it}) with a unit of number of states/(cm^2-eV). A typical distribution of D_{it}, as shown in Fig. 10.10, consists of two high peaks near the band edges and a minimum near the middle of the bandgap. The midgap levels are the most important for device operation and good processing can produce devices with minimum D_{it} of $\approx 10^{10}/(cm^2\text{-eV})$. Depending on the oxidation conditions and substrate crystal orientation, these states can either be donor-like or acceptor-like in behaviour. A donor-like state is neutral when occupied and positively charged when empty, while an acceptor-like state is negatively charged when occupied and neutral when empty. Therefore, these states give rise to surface charges, called **interface-trapped charges (Q_{it})**. Q_{it} will be negative for acceptor-like states and positive for donor-like states.

When the silicon surface layer is converted to SiO_2 by a process of oxidation, there exists a thin interfacial layer of about 50Å of composition SiO_x, where $0 < x < 2$. The excess silicon ions in this layer result in a net immobile positive charge. This charge **(Q_f)**, called the **fixed oxide charge** is not dependent on bias. It can be considered as a

constant sheet of positive charge at the Si-SiO_2 interface. Typical values of the number of charge centres per unit area, i.e. Q_f/q are $10^{10}/cm^2$ for <100> silicon and $5x10^{10}/cm^2$ for <111> silicon.

The **mobile ionic charges (Q_m),** which are due to contamination by alkali metal ions, cause reliability problems in MOS devices. They move back and forth in the oxide as the bias is changed, thereby altering the device parameters. Hence, care must be taken to eliminate the mobile ions during device fabrication.

When the device is subjected to high energy ionizing radiation or ion bombardment, charges may be trapped in the oxide. These trapped charges are called **oxide trapped charges (Q_{ot})**. These traps can be distributed throughout the oxide layer.

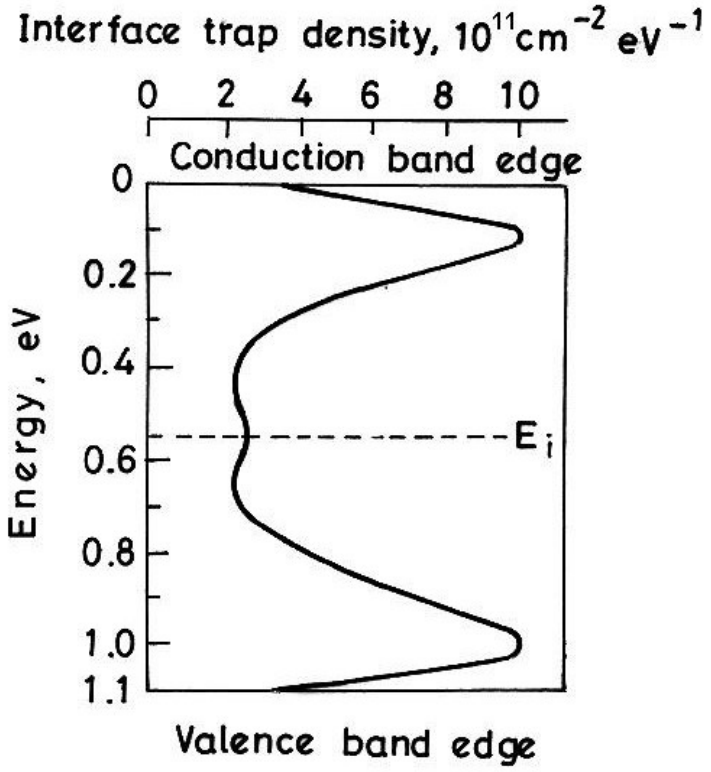


Fig. 10.10 A typical distribution of D_{it} in the bandgap of silicon[2]

Effect of Q_f, Q_m and Q_{ot}

Let us consider a MOS diode having an oxide thickness of t_{ox} and a sheet of positive charge of density Q_{ox}/cm^2 located at a distance x_o from the metal-SiO_2 interface as shown in Fig. 10.11(a). Let us assume $\phi_{ms} = 0$. Since, the device should be overall neutral, i.e. $Q_G + Q_{ox} + Q_s = 0$, negative charges will be induced in the gate metal and semiconductor. Fig. 10.11(a) also shows the electric field distribution (ε) obtained by applying Gauss' law, the potential distribution obtained by integrating ε and the corresponding band diagram. To obtain flat-band condition, a negative voltage (V_{FB}) must now be applied to the metal making Q_G more negative, so that when $Q_G = -Q_{ox}$, $Q_S = 0$. Again, the charge, electric field and potential distributions as well as the band diagram under flat-band condition, i.e. $V_{GB} = V_{FB}$ are shown in Fig. 10.11(b). From Fig. 10.11(b) we have

$$V_{FB} = E_{Fs} - E_{Fm} = -\varepsilon_o x_o = -\frac{Q_{ox}}{\varepsilon_{ox}} x_o = -\frac{Q_{ox}}{C_{ox}} \frac{x_o}{t_{ox}} \quad \text{.....(10.22)}$$

Eqn. (10.22) shows that the effect of the oxide charges on V_{FB} is dependent on the location of these charges. When they are located at the metal-semiconductor interface, i.e. $x_0 = 0$, we have $V_{FB} = 0$. The maximum influence of these charges occur when they are located at the oxide-semiconductor interface, i.e. $x_0 = t_{ox}$. Since the fixed oxide charges are always located at the Si-SiO_2 interface, the flat-band voltage due to a fixed oxide charge density of Q_f/cm^2 can be obtained by substituting $x_0 = t_{ox}$ in eqn. (10.22), so that we have

$$V_{FB} = -\frac{Q_f}{C_{ox}} \quad \text{.....(10.23)}$$

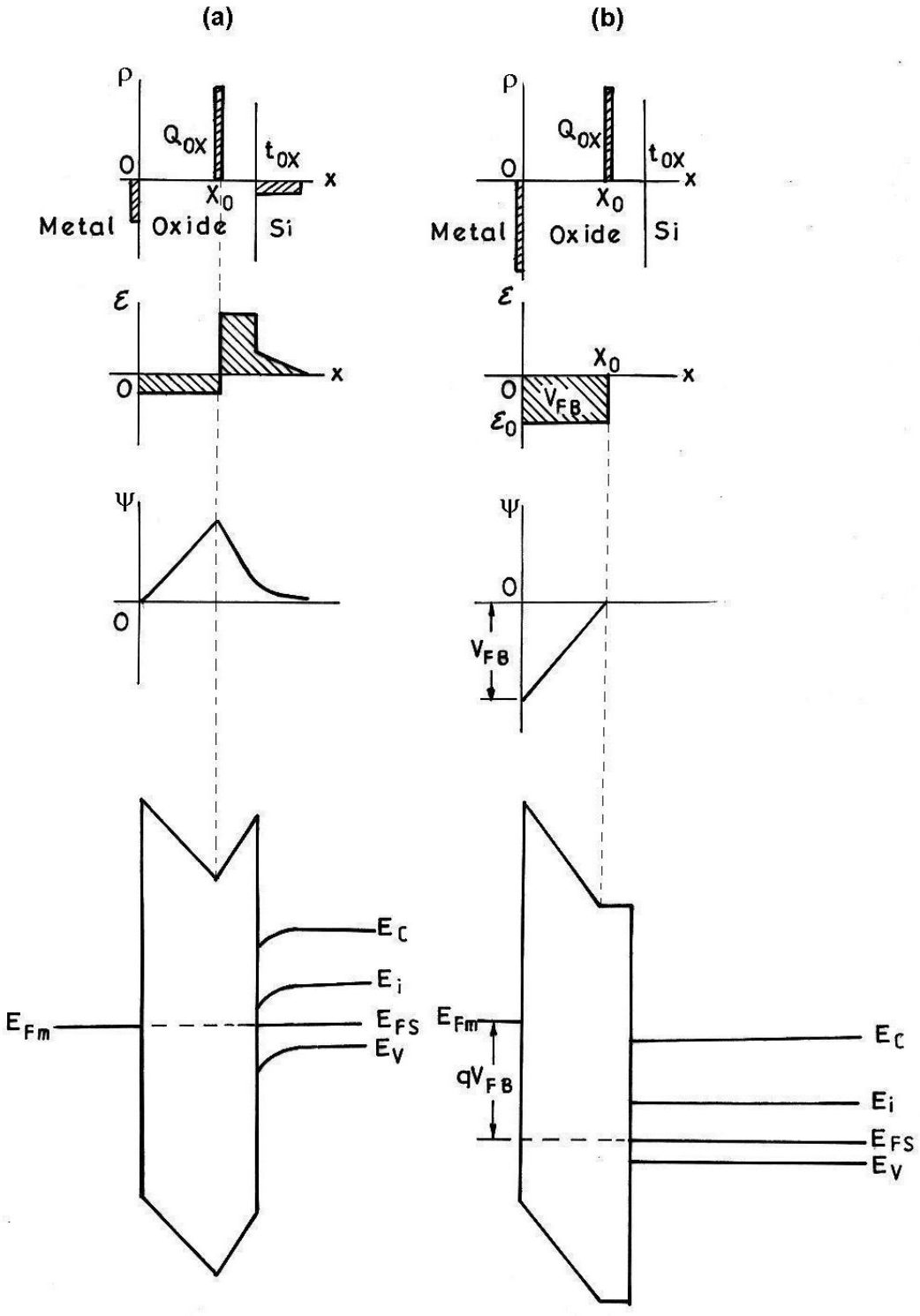


Fig. 10.11 The charge, electric field, potential variation and band diagram at (a) V_{GB} = 0V and (b)$V_{GB} = V_{FB}$ for a MOS diode having an oxide thickness of t_{ox} and a sheet of positive charge of density Q_{ox}/cm^2 located at a distance x_o from the metal-SiO_2 interface

The mobile ionic charges and oxide-trapped charges can however be distributed throughout the oxide layer. Let $\rho(x)$ be the density of charges in the charge sheet located at a distance x from the metal-SiO_2 interface. V_{FB} can now be calculated by adding the effects of the individual charge sheets in the oxide. So, from eqn.(10.22), by replacing Q_{ox} by $\rho(x)$ and integrating from 0 to t_{ox}, we have

$$V_{FB} = -\int_0^{t_{ox}} \frac{\rho(x)}{C_{ox}} \frac{x}{t_{ox}} dx = -\frac{1}{C_{ox}} \left[\frac{1}{t_{ox}} \int_0^{t_{ox}} x\,\rho(x) dx \right] \text{.....(10.24)}$$

Let $\rho_m(x)$ and $\rho_{ot}(x)$ represent the distributions of the mobile ionic charges and oxide-trapped charges respectively. We now define the terms Q_m and Q_{ot} as

$$Q_m = \frac{1}{t_{ox}} \int_0^{t_{ox}} x\rho_m(x) dx \text{ .....(10.25a)}$$

$$Q_{ot} = \frac{1}{t_{ox}} \int_0^{t_{ox}} x\rho_{ot}(x) dx \text{ .....(10.25b)}$$

Now, for a MOS diode with $\phi_{ms} \neq 0$, and non-zero values of Q_f, Q_m and Q_{ox}, we have from eqns. (10.19), (10.23),(10.24) and (10.25)

$$V_{FB} = \phi_{ms} - \frac{Q_f + Q_m + Q_{ot}}{C_{ox}} \text{ .....(10.26)}$$

Note that Q_m and Q_{ot} do not represent the total mobile and oxide trapped charges, respectively, but an equivalent charge, which when located at the Si-SiO_2 interface would have the same effect as the actual charge distribution. Eqn. (10.26) is a general expression for the flat-band voltage, and the threshold voltage can be calculated by substituting the value of V_{FB} in eqn.(10.21).

The effect of Q_{it} is more complicated and its effect on the C-V characteristics will be taken up at the end of the next section.

Example 10.5. The MOS diode MD4 is similar to MD3, except that MD4 has a fixed oxide charge density given by $Q_f/q = 8x10^{10}/cm^2$, while $Q_m=Q_{ot}=0$. Calculate the threshold voltage.

For MD3, $C_{ox} = 3.45x10^{-8}F/cm^2$, $(V_{th})_{ideal} = 2.11V$ (Ex. 10.3), $\phi_{ms} = - 0.9V$(Ex. 10.4)

From eqn.(10.26), $V_{FB} = -0.9 - \frac{1.6x10^{-19} x8x10^{10}}{3.45x10^{-8}} = -1.27V$

Therefore, $V_{th} = V_{FB} + (V_{th})_{ideal} = -1.27 + 2.11 = 0.84V$

Example 10.6. A MOS diode MD5 having a threshold voltage of 1V is to be fabricated. Just like MD3, MD5 has n^+ polysilicon gate. Its substrate doping concentration is similar to MD1 and MD3, i.e. $N_A=1x10^{16}/cm^3$. It has a fixed oxide charge density given by $Q_f/q=8x10^{10}/cm^2$ and $Q_m=Q_{ot}=0$. Calculate the gate oxide thickness (t_{ox}) required to achieve $V_{th}=1V$.

For MD1, $\phi_B = 0.35V$and $W_{max} = 0.304\mu m$ (Ex. 10.3) and for MD3, $\phi_{ms}= - 0.9V$(Ex. 9.4)

Hence, from eqns. (10.21), (10.26) and (10.15)

$$1 = -0.9 - \frac{1.6x10^{-19} x8x10^{10}}{C_{ox}} + 2x0.35 + \frac{1.6x10^{-19} x10^{16} x0.304x10^{-4}}{C_{ox}}$$

or, $C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}} = 3x10^{-8}\ F/cm^2$

Therefore, $t_{ox} = \frac{3.9x8.85x10^{-14}}{3x10^{-8}} = 1.15x10^{-5}\,cm = 1150Å$

10.2.4 C-V Characteristics of the MOS diode (capacitor)

We cannot use the current-voltage (I-V) characteristics to characterize the MOS diode, since no current flows through the diode in the steady state. It can, however, be characterized as a capacitor. Hence, the MOS diode is mostly referred to as a MOS capacitor. The capacitance-voltage (C-V) characteristics can be used to obtain a lot of information about the device. A typical measurement setup is shown in Fig. 10.12. A dc bias (V_{GB}) is applied to the MOS capacitor. The capacitance meter measures the capacitance by applying a small ac voltage (typically 50mV) on top of the dc bias and measuring the reactive current. By changing the dc bias using a ramp generator and plotting the capacitance as a function of dc bias, the C-V plot is obtained.

C-V plot for $Q_{it} = 0$

In a MOS capacitor, $Q_G + Q_{ox} + Q_S = 0$, since the MOS capacitor should be overall charge neutral. Assuming Q_{ox} is independent of V_{GB}, the small signal capacitance per unit area is given by

$$C = \frac{dQ_G}{dV_{GB}} = -\frac{dQ_S}{dV_{GB}} \quad \text{.....(10.27)}$$

Hence, $$\frac{1}{C} = -\frac{dV_{GB}}{dQ_S} = -\frac{d}{dQ_S}\left[V_{FB} + \psi_s - \frac{Q_S}{C_{ox}}\right] = \frac{1}{C_S} + \frac{1}{C_{ox}} \quad \text{.....(10.28)}$$

where $C_S = -\frac{dQ_S}{d\psi_S}$ and $C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}}$. So the equivalent circuit of the MOS capacitor as shown in Fig.10.13 consists of two capacitors, C_S and C_{ox} in series. Of the two capacitors, C_{ox} is a constant for a particular MOS capacitor while C_S is dependent on bias. We shall now discuss the nature of the C-V plot in the different regions of operation.

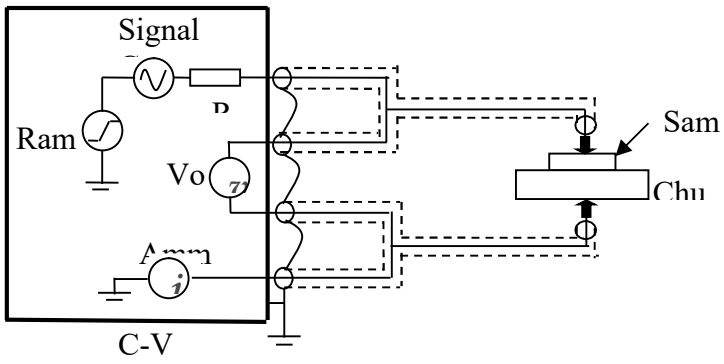


Fig. 10.12 A typical setup to measure the capacitance-voltage (C-V) characteristics of MOS capacitors

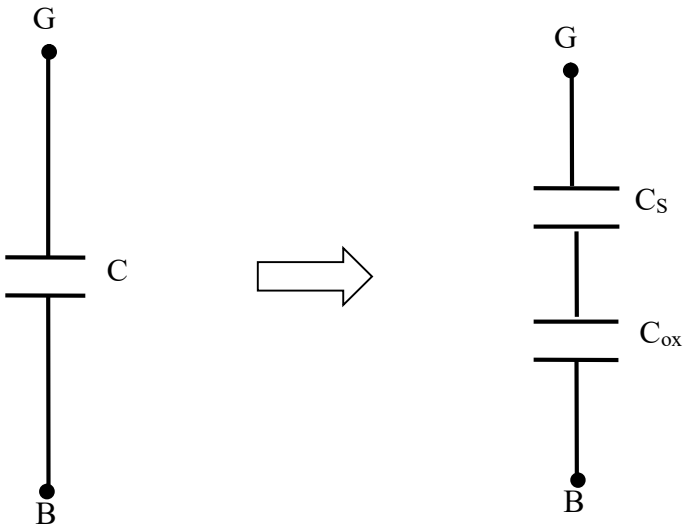


Fig. 10.13 Equivalent circuit of the MOS capacitor

(a) **Accumulation region of operation**: $V_{GB} < V_{FB}$, $\psi_s < 0$

In the accumulation condition, the hole concentration at the semiconductor surface depends exponentially on ψ_s. This implies that a small change in ψ_s will result in a large increase in Q_S. In other words, C_S will be very large. From eqn.(10.28), if $C_S >> C_{ox}$, $C = C_{ox}$. Physically, the result can be explained by noting that the high density of charge carriers in the accumulation layer makes the semiconductor surface behave almost like a metal plate. Thus, in the accumulation region, the MOS capacitor is electrically equivalent to a capacitor with two metal plates separated by a distance t_{ox}.

(b) **Depletion region of operation**: $V_{FB} < V_{GB} < V_{th}$, $0 < \psi_s \leq 2\phi_B$

In the depletion condition, the semiconductor charge is composed mostly of the uncovered impurity acceptor ions and is given by eqn.(10.10). Therefore we have

$$C_S = -\frac{dQ_S}{d\psi_S} = \sqrt{\frac{qN_A\varepsilon_s}{2\psi_S}} = \varepsilon_s\sqrt{\frac{qN_A}{2\varepsilon_s\psi_S}} = \frac{\varepsilon_s}{W} \text{ ……..(10.29)}$$

Thus, the capacitance is the same as that of a pn junction width depletion width W or a capacitor having two metal plates separated by a distance W. Hence, from eqns.(10.28) and (10.29), we have

$$\frac{1}{C} = \frac{1}{C_{ox}} + \frac{W}{\varepsilon_s} \text{ ……(10.30)}$$

Now, as V_{GB} increases, W increases, so C reduces. At threshold voltage, i.e. $V_{GB} = V_{th}$, $W = W_{max}$, and the depletion region stops expanding. Therefore the capacitance per unit area reaches a minimum value (C_{min}) at $V_{GB} = V_{th}$ given by

$$\frac{1}{C_{min}} = \frac{1}{C_{ox}} + \frac{W_{max}}{\varepsilon_s} \text{ ……..(10.31)}$$

The variation of capacitance with applied voltage is shown in Fig. 10.14.

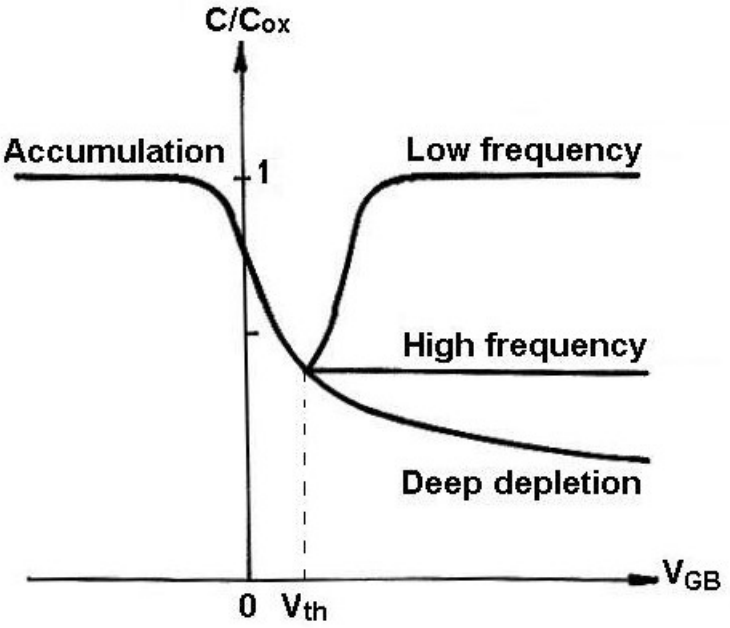


Fig. 10.14 The C-V characteristics of MOS capacitors showing the measurements at low frequency(LF), high frequency (HF) and deep depletion (DD) conditions

Example 10.7. If the device MD1 has an area (A) of $1x10^{-4}$ cm^2, what is its (i) maximum capacitance (C'_{max}), (ii) minimum capacitance (C'_{min})? [ε_r = 11.9 for Si]

For device MD1, W_{max} =0.304μm, C_{ox} = $3.45x10^{-8}$ F/cm^2 (from Ex. 9.3).

(i) $C'_{max} = A\ C_{ox} = 1x10^{-4} \times 3.45x10^{-8} = 3.45x10^{-12}$ F = 3.45 pF.

(ii) From eqn.(10.31), $\frac{1}{C_{min}} = \frac{1}{3.45x10^{-8}} + \frac{0.304x10^{-4}}{11.9x8.85x10^{-14}}$

Therefore $C_{min} = 1.73x10^{-8}$ F/cm^2 and $C'_{min} = 1x10^{-4} \times 1.73x10^{-8} = 1.73x10^{-12}$F = 1.73 pF.

Q. 10.3. How does (C_{max}/C_{min}) ratio depend on N_A and t_{ox} ?

From eqn.(10.31), $\frac{C_{max}}{C_{min}} = \frac{C_{ox}}{C_{min}} = 1 + \frac{C_{ox} W_{max}}{\varepsilon_s}$

W_{max} decreases with increase in N_A. Also, with increase in t_{ox}, C_{ox} decreases. Hence increase in either N_A or t_{ox} results in a decrease in the (C_{max}/C_{min}) ratio.

It is often useful to have an expression showing the variation of C with V_{GB}. Substituting the values of ψ_s and Q_S from eqns.(10.9) and (10.10) in eqn.(10.20), we have a quadratic equation in W given by

$$V_{GB} - V_{FB} = \frac{qN_A W^2}{2\varepsilon_s} + \frac{qN_A W}{C_{ox}} \quad \text{.....(10.32)}$$

The solution of this quadratic equation is of the form

$$W = \frac{\varepsilon_{Si}}{C_{ox}} \left[\sqrt{1 + \frac{2C_{ox}^2 (V_{GB} - V_{FB})}{qN_A \varepsilon_s}} - 1 \right] \text{.....(10.33)}$$

Substituting the above expression for W in eqn.(10.30) we have

$$C = \frac{C_{ox}}{\sqrt{1 + \frac{2C_{ox}^2 (V_{GB} - V_{FB})}{qN_A \varepsilon_s}}} \text{.....(10.34)}$$

The above expression is often used to extract the doping concentration of the semiconductor substrate (N_A) from C-V measurements. Eqn.(10.34) can be rewritten as

$$\frac{1}{C'^2} = \frac{1}{C'^2_{ox}} + \frac{2 (V_{GB} - V_{FB})}{A^2 qN_A \varepsilon_s} \text{.....(10.35)}$$

where A is the area of the diode and C'=AC is the actual measured capacitance. If we now plot the variation of $(1/C'^2)$ versus V_{GB}, we will get a straight line whose slope is equal to $2/(A^2qN_A\varepsilon_s)$. This is similar to the case already discussed in Section 4.2.1 for an abrupt n^+p junction. Fig. 10.15 shows the plots for two diodes having different substrate doping concentrations. If the area of the device is known, N_A can easily be calculated from the slope of the line.

A point to note is the effect of V_{FB} on the C-V characteristics. From eqn.(10.34), we see that C depends on $(V_{GB}-V_{FB})$, i.e. if we have two diodes with identical N_A and C_{ox} but with different flat-band voltages, the capacitance per unit area will be the same at the same $(V_{GB}-V_{FB})$. Suppose the capacitance per unit area of an ideal MOS diode (i.e. V_{FB} = 0V) at a voltage V_{GB1} is C_1. Then a MOS diode having identical N_A and C_{ox} will have the same capacitance per unit area (C_1) at an applied voltage of $V_{GB1}+V_{FB}$, since the value of $V_{GB}-V_{FB}$ for both the devices is V_{GB1}. This implies that the C-V characteristics of a diode with $V_{FB} \neq 0V$ will experience a parallel shift of V_{FB} with the respect to the corresponding ideal diode curve. The C-V characteristics of two such diodes are shown in Fig.10.16.

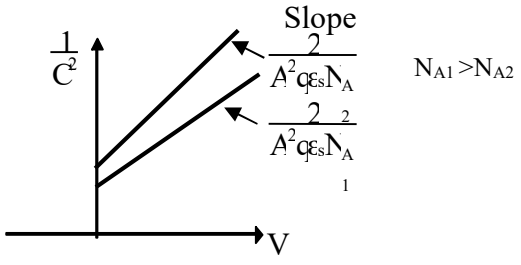


Fig. 10.15 $(1/C'^2)$ versus V_{GB} plots for two diodes having different substrate doping concentrations.

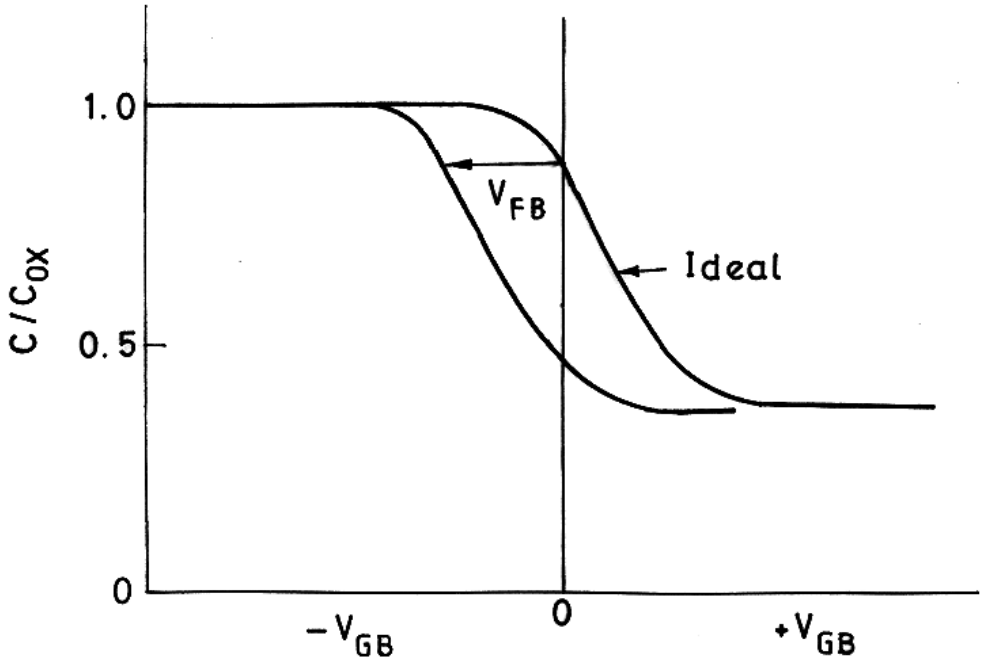


Fig. 10.16 The C-V characteristics of diodes with $V_{FB} = 0V$ and $V_{FB} \neq 0V$.

(c) **Strong inversion region of operation**: $V_{GB} > V_{th}$, $\psi_s \geq 2\phi_B$

For $V_{GB} > V_{th}$, $Q_S = Q_B + Q_n$ and Q_n is comparable to Q_B. In this region of operation, any increase in Q_S is due to the increase in Q_n, since the depletion width has reached its maximum value, making Q_B constant. So it is expected that C_S is going to be extremely large since Q_n increases exponentially with ψ_s. This is true, provided ψ_s and V_{GB} change slowly. However, electrons in the surface inversion layer cannot respond to rapid change in V_{GB}, and in particular to the high frequency (~1MHz) signal used to measure the capacitance. This is because any addition or decay in the surface concentration of electrons, which are minority carriers in the semiconductor bulk, is due to the process of thermal generation or recombination. Unless the minority carrier lifetime is abnormally small, these generation/recombination processes are extremely slow. For example, in a MOS capacitor fabricated on a high quality silicon substrate, it

may take a few seconds for the inversion layer to form after it is biased in the strong inversion condition. (This problem does not exist in a MOSFET, where electrons can be drawn from the n^+ doped source region). So we see, that depending on the measurement conditions, we may get different C-V plots in the strong inversion region as follows:

(i) A low frequency signal (few Hertz) is used to measure the capacitance and V_{GB} is changed slowly: Due to the a.c. signal on the gate, the semiconductor charge should also vary at the same rate. In this case, the electron concentration in the inversion layer is able to change in step with the applied small signal. Since C_S is extremely large, from eqn.(10.28), $C = C_{ox}$. So the capacitance increases to the maximum value as shown in the LF plot in Fig. 10.14.

(ii) A high frequency signal (~1MHz) is used to measure the capacitance and V_{GB} is changed slowly: As a high frequency a.c. signal is applied on the gate, the electron concentration in the inversion layer is unable to change in step with the applied small signal. The depletion width, however, widens and narrows at the same rate as the a.c. signal. Since the capacitance is now due to variation of the depletion width, it is given by eqn. (10.30), where $W = W_{max}$ is the average depletion width. So the capacitance remains constant at C_{min} given by eqn.(10.31) as shown in the HF plot in Fig. 10.14.

(iii) V_{GB} is changed quickly (about 10V/s and higher): In this case, the inversion layer does not get sufficient time to form. But as V_{GB} is increased, the positive charge on the metal keeps increasing and there must be an equal increase in the negative charge in the semiconductor. To achieve this, the depletion width keeps increasing beyond W_{max} and the capacitance which is given by eqn.(10.28) is less than the

value of C_{min} given by eqn.(10.31). This is called the **deep depletion condition** and the C-V plot for this condition is also shown in Fig.10.14. It must be remembered that deep depletion is a non-equilibrium condition. If the voltage sweep is stopped suddenly in this condition, the inversion layer will slowly form due to generation of electrons and the capacitance will approach C_{min}.

In Fig. 10.17, the solid line shows the plots obtained from the equations presented above, while the dotted line shows the nature of the plots obtained experimentally. The deviation is due to the fact that we have used the depletion approximation in our analysis, thereby neglecting the carrier concentrations in the depletion region. However, the hole concentration is significant near flat-band condition and the electron concentration near threshold. This results in the experimental C-V curves being smoothened out in these regions.

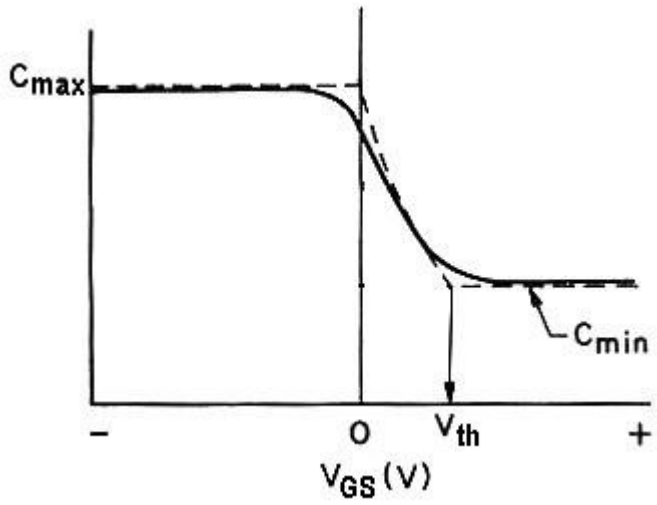


Fig. 10.17 Comparison of the nature of C-V plots obtained from our analysis (dashed line) with those obtained experimentally (solid line)

C-V plot for $Q_{it} \neq 0$

We have already seen that the C-V characteristics of a MOS capacitor is shifted by V_{FB} when compared to the characteristics of the corresponding ideal device (Fig. 10.16). When the fixed oxide charge density is Q_f/cm^2, the resultant shift due to it is equal to $-Q_f/C_{ox}$. The presence of interface trapped charge (Q_{it}) has a similar effect, giving rise to a shift of $-Q_{it}/C_{ox}$. However, there are certain differences. While Q_f is always positive, Q_{it} can be either positive or negative, depending on whether the states are donor-like or acceptor-like. Also Q_f is fixed and does not depend on bias, so that it results in a parallel shift from the ideal C-V curve. On the other hand, Q_{it} changes with bias, and hence the shift from the ideal C-V curve changes continuously from the accumulation to the inversion condition. This can be clearly understood from Fig. 10.18, which shows a p-type silicon substrate with acceptor-like states distributed at the surface. We assume that all states below the Fermi level are filled giving rise to negative charges, while all the states above the Fermi level are empty and therefore neutral. As shown in Fig.10.18(a), when the silicon surface is in accumulation, some states near the valence band lie under the Fermi level, giving rise to a small amount of negative charges. As we move towards inversion, the bands bend downwards, and more and more states are filled as shown in Fig. 10.18(b). The resulting C-V curve (Fig. 10.19) shows a very small positive shift in accumulation, gradually increasing to a maximum value in inversion. The threshold voltage of the MOS device is increased as the additional positive charge on the gate is balanced by increasing Q_{it} rather than increasing Q_S. This is one of the reasons why it is almost impossible to create an inversion layer in GaAs, which has a large density of acceptor-like surface states.

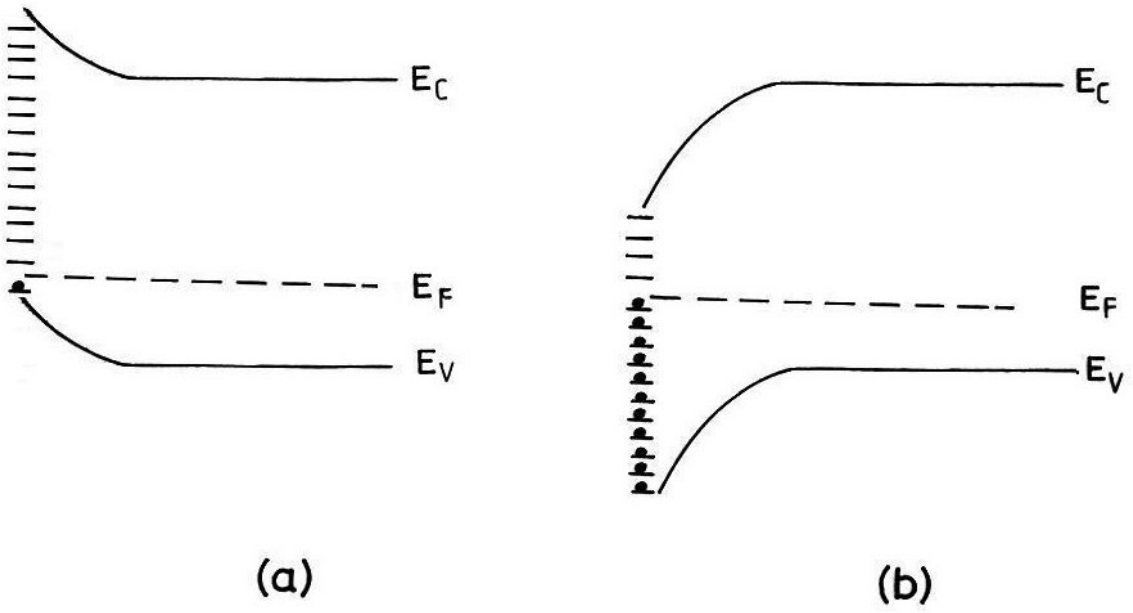


Fig. 10.18 Change in charge of acceptor-like surface states due to band bending

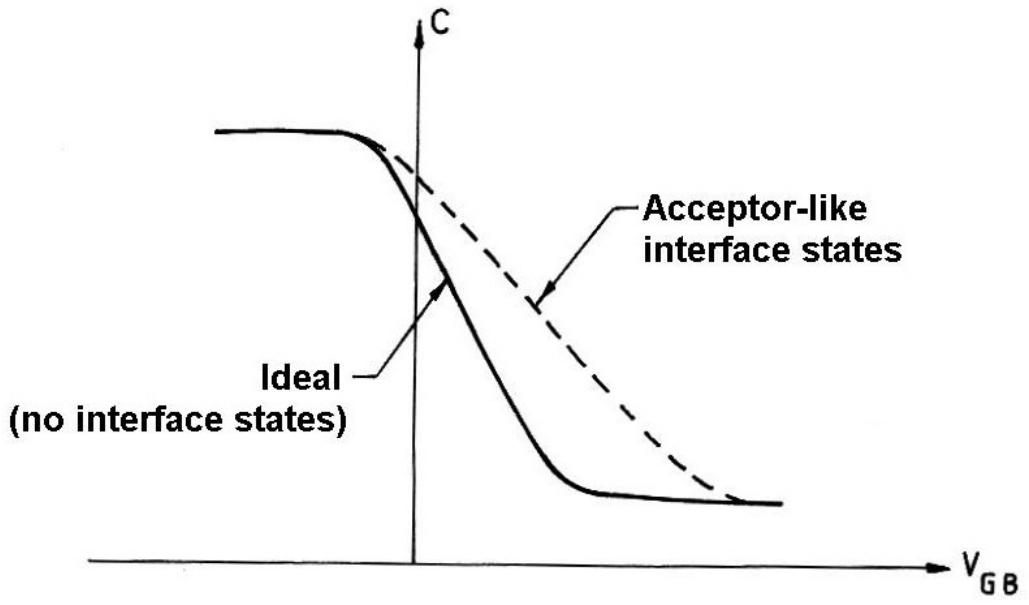
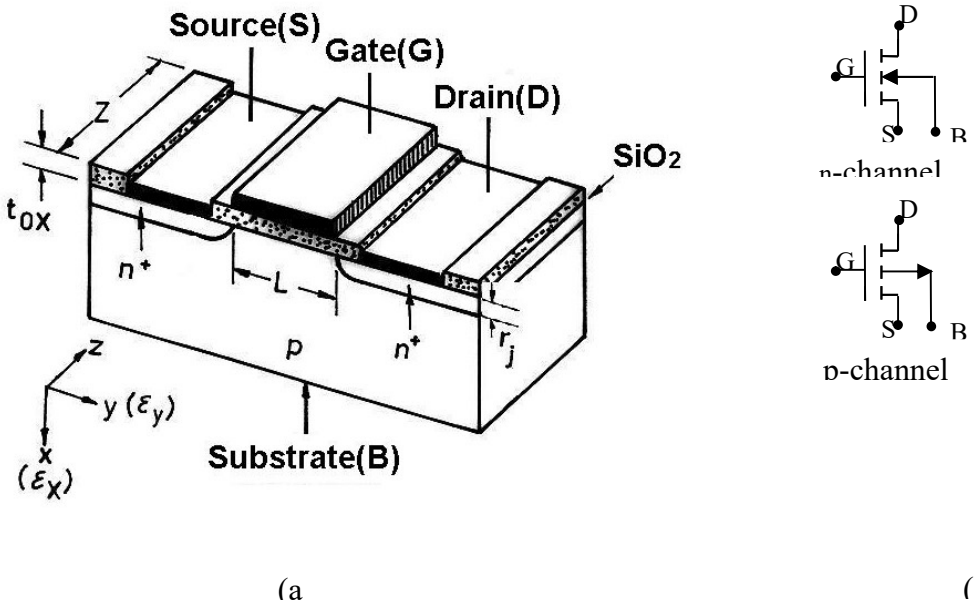


Fig. 10.19 Shift in CV plot due to surface states



(a (

Fig. 10.20 (a) Perspective view of a four-terminal n-channel MOSFETand (b) symbols for n-channel and p-channel MOSFETs

10.3 The MOSFET

Fig. 10.20 shows a perspective view of a four-terminal n-channel MOSFET and the symbols for n-channel and p-channel MOSFETs. In addition to the MOS structure, the MOSFET consists of n^+ diffused source and drain regions. The space separating the source and drain is called the channel of the MOSFET. Fig. 10.20 also shows the important parameters of the MOSFET, viz. the **channel length 'L'**, which is the distance between the two pn junctions, the **channel width 'Z'**, the **gate oxide thickness 't_{ox}'**, the **source and drain junction depths 'r_j'** and the **substrate doping concentration 'N_A'**. In this section, we shall use these parameters to develop models of the n- channel MOSFET. A p-channel MOSFET has p^+ doped source and drain regions and an n-type substrate. Although, all the analyses in this chapter are for n-channel MOSFETs, they can also be applied to p-channel MOSFETs by changing the polarity of the terminal voltages. In

MOSFETs, the terminal voltages are usually referred to with respect to the source. We shall be following the same convention in this chapter, i.e. the model equations will be developed with respect to V_{GS}, V_{DS} and V_{BS}, which are the gate-to-source voltage, the drain-to-source voltage and the bulk-to-source voltage respectively. For the normal operation of an n-channel MOSFET, V_{GS} and V_{DS} are positive voltages, while V_{BS} is always negative or zero.

Let us at first discuss the operation of the MOSFET qualitatively. The MOSFET consists of two p-n junctions connected back to back. We have already seen in the discussion for a MOS diode, that when the gate-to-bulk voltage exceeds the threshold voltage (i.e. $V_{GB} > V_{th}$), an inversion layer is formed. An inversion layer is also formed in the MOSFET channel, when its gate-to-source voltage exceeds its threshold voltage (i.e. $V_{GS} > V_{th}$). It may be noted that the threshold voltage for a MOSFET need not be the same as the value for the corresponding MOS diode, as in one case it is defined as a gate-to source voltage while in the other it is a gate-to-bulk voltage. When $V_{GS} < V_{th}$, the MOSFET channel is either accumulated or depleted and there is no conducting path between the source and drain. When accumulation condition exists in the channel, the only current which can flow is the reverse leakage current of the p-n junction diodes. When the channel region is depleted, a diffusion current exists, with electrons flowing from the source to drain under the influence of a concentration gradient of electrons. This current, called the subthreshold current, reduces exponentially as V_{GS} is reduced below V_{th}. The subthreshold current will be discussed in detail in the next chapter.

When $V_{GS} > V_{th}$, the silicon surface is inverted and an n-type conducting channel exists between the n^+ source and drain regions. Now if a positive voltage is applied to the drain

with respect to the source (i.e. $V_{DS} > 0$), electrons flow from the source towards the drain under the influence of the electric field. The above-threshold current in MOSFETs is therefore primarily a drift current. The inversion layer behaves as a resistance, and as V_{DS} is increased initially from 0V, the drain current (I_D) increases linearly. This is shown in Fig. 10.21(a). With increasing V_{DS}, however, the current-voltage (I-V) characteristics deviate from linearity. This is because, with increasing V_{DS} and a fixed V_{GS}, the voltage across the oxide and consequently the electron concentration reduces from the source to the drain end of the channel. Ultimately, when $V_{DS} = V_{Dsat}$, the voltage across the oxide at the drain end is no longer sufficient to support an inversion layer. This is called the **pinch-off point** and is shown in Fig. 10.21(b). Since the voltage across the oxide determines the electron concentration in the semiconductor, V_{Dsat} will be higher for higher V_{GS}. When $V_{DS} < V_{Dsat}$, the device is said to be in the **linear region** of operation. For $V_{DS} > V_{Dsat}$, the drain current remains almost constant. In this region, called the **saturation region** of the characteristics, the pinch-off point moves towards the source as V_{DS} increases. The voltage at the pinch-off point remains V_{Dsat}, the extra voltage (V_{DS} - V_{Dsat}) being now dropped in a depletion layer. Electrons, which reach the pinch-off point, are swept across to the drain by the electric field in the depletion region. This is quite similar to the way injected carriers are swept across the reverse biased base-collector junction in a bipolar transistor. The current in the saturation region remains almost fixed as the voltage across the inversion layer is fixed at V_{Dsat}. A small increase in current is observed due to the shift of the pinch-off point towards the source end, which effectively reduces the channel length of the MOSFET from L to L' as shown in Fig. 10.21(c).

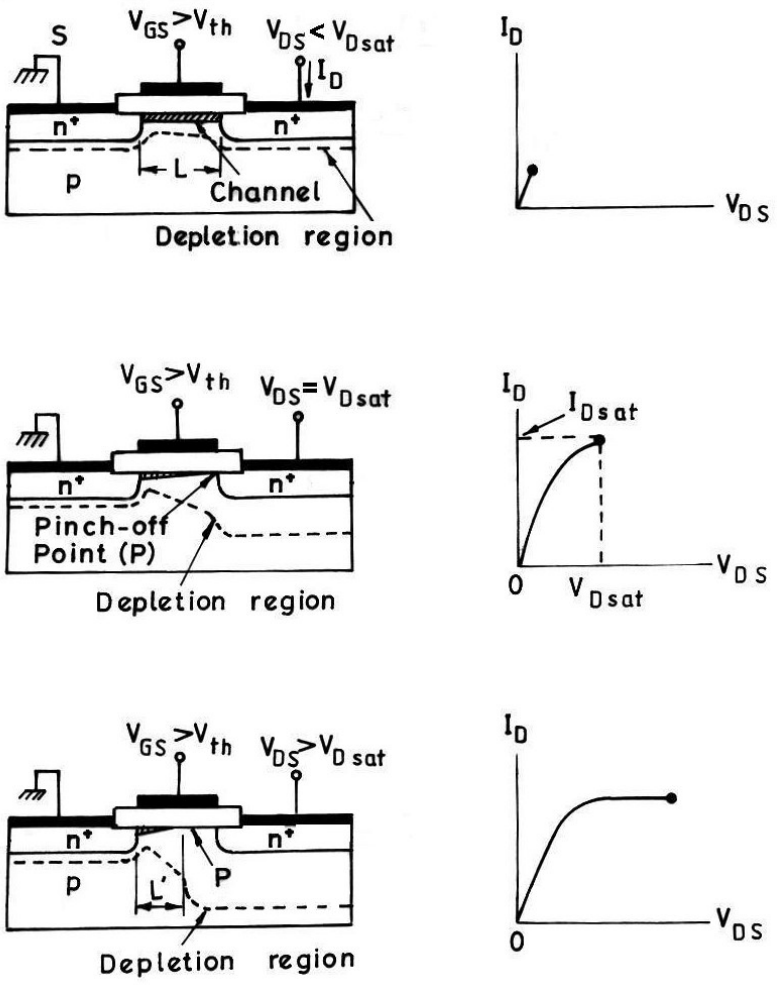


Fig. 10.21 Various regions of operation of a MOSFET

The above discussion was for a particular value of $V_{GS}>V_{th}$. With further increase in V_{GS}, the channel conductivity increases, resulting in an increase in I_D for all values of V_{DS}. The complete I-V characteristics of a MOSFET is shown in Fig. 10.22.

We shall now proceed to develop analytical expressions for the I-V characteristics of MOSFETs. Before taking up the derivation of expressions for the drain current, we shall discuss about the threshold voltage (V_{th}), which is one of the most important parameters of a MOSFET.

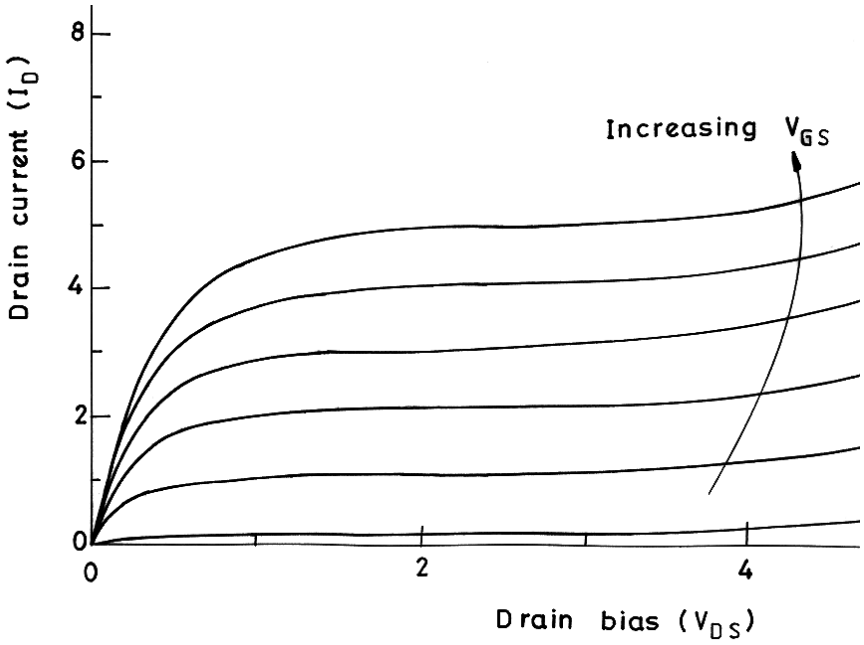


Fig. 10.22 I-V characteristics of a MOSFET

10.3.1 Threshold voltage of MOSFET

In a MOSFET, since all voltages are measured with respect to the source, the threshold voltage is measured as a particular value of gate-to-source voltage and not a gate-to-bulk voltage as in a MOS diode. **The threshold voltage (V_{th}) of a MOSFET is defined as the value of the gate-to-source voltage which is just sufficient to produce a surface inversion layer when V_{DS} = 0V.**

Due to the application of a substrate bias with respect to the source and drain, the channel region is no longer in thermal equilibrium. For an n-channel MOSFET, V_{BS} is usually less than or equal to zero, i.e. the source and drain junctions are reverse biased, to avoid the flow of a substrate current. We have seen earlier in Section 4.3.2 that in such a situation, the Fermi level splits into the electron and hole quasi Fermi levels separated by an energy (in eV) equal to the applied bias across the p-n junctions (in V). For a reverse

bias of V_R, the electron quasi Fermi level (E_{Fn}) moves down in the p-side, and at the edge of the depletion region, $(E_F - E_{Fn}) = qV_R$, where E_F is the Fermi level in the bulk. In a MOSFET, considering the p-n junctions at the source and drain, we have $(E_F - E_{Fn}) = -qV_{BS}$ and $(E_F - E_{Fn}) = q(V_{DS} - V_{BS})$ at the source and drain ends of the channel respectively. There is therefore a continuous variation in E_{Fn} from the source to the drain end of the channel when $V_{DS} \neq 0$, which accounts for the flow of electrons.

The threshold voltage is, however, defined for $V_{DS} = 0V$. For this case, the position of E_{Fn} is constant throughout the channel and is given by $E_{Fn} = E_F + qV_{BS}$. The band diagram of the MOS structure in the channel region at the threshold condition is shown in Fig. 10.23. Now, at the onset of strong inversion, the electron concentration at the semiconductor surface is equal to the hole concentration in the bulk, that is

$$n_{ps} = N_A = n_i \exp\left(\frac{E_{Fn} - E_{iS}}{kT}\right) \ldots\ldots\ldots\ldots\ldots\ldots (10.36)$$

where E_{iS} is the intrinsic level at the semiconductor surface, which is related to the intrinsic level at the bulk (E_i) through the relation $E_{iS} = E_i - q\psi_S$. Now, expressing E_{Fn} and E_{iS} in eqn.(10.36) in terms of E_F and E_i, we have

$$N_A = n_i \exp\left(\frac{E_F + qV_{BS} - E_i + q\psi_S}{kT}\right)$$

$$\text{or, } E_F - E_i + q(V_{BS} + \psi_S) = kT \ln (N_A/n_i) = q\phi_B$$

Since, $q\phi_B = E_i - E_F$, we have at threshold condition

$$\psi_S = 2\phi_B - V_{BS} \ldots\ldots\ldots\ldots\ldots\ldots (10.37)$$

This implies that for $V_{BS} < 0V$, the potential drop in the semiconductor at threshold is greater than in a MOS capacitor, resulting in an increase in threshold voltage.

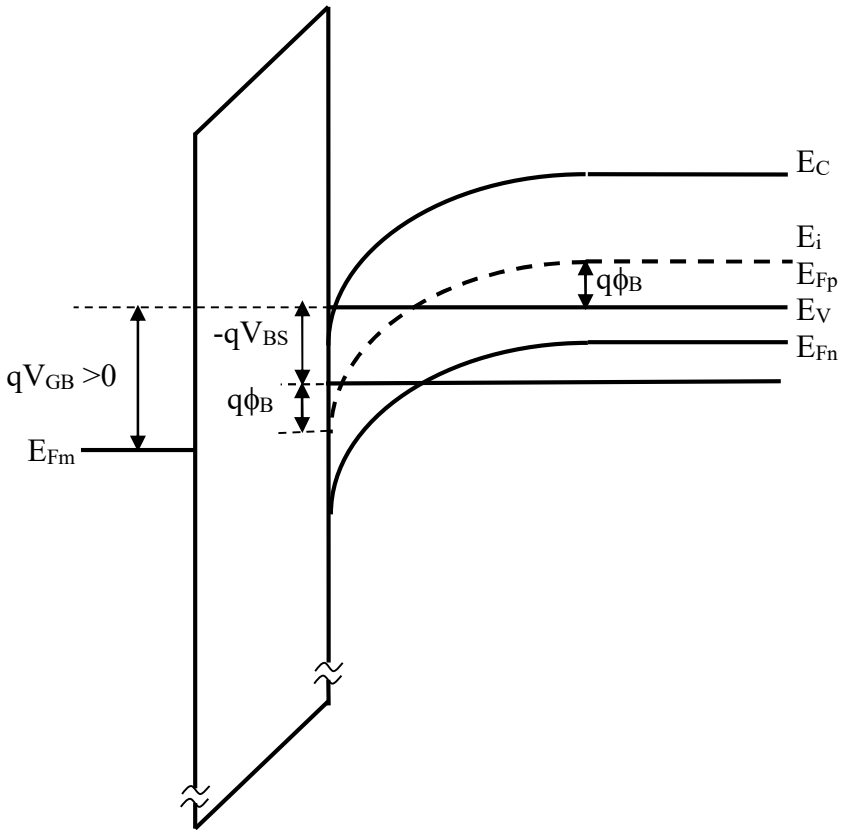


Fig.10.23 Band diagram in the channel region of a MOSFET at threshold condition

Assuming that Q_n is negligible at threshold, so that $Q_S \approx Q_B$, we have from eqn. (10.20)

$$V_{GB} - V_{FB} = \psi_S - \frac{Q_B}{C_{ox}} = 2\phi_B - V_{BS} + \frac{\sqrt{2\varepsilon_S qN_A(2\phi_B - V_{BS})}}{C_{ox}} \quad \text{.....(10.38)}$$

Now, $V_{GS} = V_{GB} + V_{BS}$, and since $V_{GS}=V_{th}$ at threshold, we have from eqn.(10.38)

$$V_{th} = V_{FB} + 2\phi_B + \frac{\sqrt{2\varepsilon_S qN_A(2\phi_B - V_{BS})}}{C_{ox}} \quad \text{.....(10.39)}$$

We see from the above equation that when V_{BS} = 0V, the threshold voltage for the MOSFET reduces to that for the corresponding MOS capacitor given by eqn. (10.21). The effect of V_{BS} on the threshold voltage can now be expressed through the relation

$$\Delta V_{th} = V_{th} - V_{tho} = \gamma\left(\sqrt{2\phi_B - V_{BS}} - \sqrt{2\phi_B}\right) \ldots\ldots\ldots\ldots (10.40)$$

where ΔV_{th} is the change in threshold voltage due to the application of substrate bias, V_{tho} is the threshold voltage for V_{BS} = 0V given by eqn.(10.21) and **γ is called the body-effect parameter** given by

$$\gamma = \frac{\sqrt{2\varepsilon_S q N_A}}{C_{ox}} \ldots\ldots\ldots..(10.41)$$

From eqn.(10.41), we see that the body-effect, which is the change in V_{th} due to a substrate bias, is more pronounced in MOSFETs with higher N_A and t_{ox}.

Example 10.8. The MOS structure in MOSFET MF1 is similar to the MOS diode MD4, i.e. it has an n^+ polysilicon gate, the gate oxide thickness (t_{ox}) is 1000Å and the substrate doping concentration is $1x10^{16}/cm^3$. Calculate the threshold voltage of MF1 for (i) V_{BS} = 0V, (ii) V_{BS} = -2V and (iii) V_{BS} = –5V.

From eqn.(9.41), $\gamma = \dfrac{\sqrt{2x11.9x8.85x10^{-14}x1.6x10^{-19}x10^{16}}}{3.45x10^{-8}} = 1.683$

The threshold voltage (V_{th}) is calculated using eqn. (10.40), where V_{tho} is the threshold voltage of the corresponding MOS capacitor. From example 10.5, V_{th} for MD5 is 0.84V. Also, ϕ_B = 0.35V for $N_A=10^{16}/cm^3$ (Example 10.1). Therefore from eqn.(10.40),

(i) when V_{BS}=0V, $V_{th} = V_{tho} = 0.84V$

(ii) when V_{BS}= -2V, $V_{th} = 0.84 + 1.683\left(\sqrt{2.7} - \sqrt{0.7}\right) = 2.2V$

(iii) when $V_{BS} = -5V$, $V_{th} = 0.84 + 1.683\left(\sqrt{5.7} - \sqrt{0.7}\right) = 3.45V$

10.3.2 Above-threshold I-V Characteristics of MOSFETs

In this section, we shall extend the analysis for MOS diodes to derive expressions for the drain current I_D of the four-terminal MOSFET as a function of its terminal voltages when $V_{GS}>V_{th}$. In order to simplify the analysis, we shall initially make some assumptions. We shall subsequently discuss the cases where these assumptions are not valid. The assumptions are: (1) There are no interface states. (2) The drift component of current is much larger than any diffusion component. (3) Carrier mobility in the channel is constant. (4) The channel region is uniformly doped. (5) The transverse electric field (E_x), which is perpendicular to the direction of current flow, is much larger than the longitudinal field (E_y) in the direction of current flow. This condition is called the **gradual channel approximation (GCA)**, and MOSFETs where this approximation is valid are called **long-channel MOSFETs**.

The two-dimensional nature of the device (i.e. the presence of two electric fields perpendicular to one another) makes it difficult to obtain simple, yet accurate analytical expressions for the current-voltage characteristics of MOSFETs. The gradual channel approximation is used to simplify the problem. Here the MOSFET channel is imagined to consist of a large number of MOS capacitors, which fill the space between the source and drain. Since the longitudinal electric field can be ignored, the one-dimensional analysis for MOS capacitors can be used to calculate the charges Q_n and Q_B at the semiconductor surface. These charges vary from capacitor to capacitor, depending on their position with

respect to the source and drain. Once the electron distribution along the channel is obtained, the drain current can be calculated by considering the longitudinal electric field through a separate one-dimensional analysis. The gradual channel approximation therefore enables us to reduce a two-dimensional problem into simpler one-dimensional form.

Several analytical models of the above threshold conduction of MOSFETs have been proposed. In this book, we shall discuss three of these models, which are available as Level 1, Level 2 and Level 3 MOSFET models in the popular circuit simulator SPICE.

10.3.2.1 Bulk charge model (Level 2 in SPICE)

Linear region characteristics

Let us at first derive a relation for the I-V characteristics of a MOSFET in the linear region of operation, i.e. when a continuous inversion layer exists from the source to the drain end of the channel. To help us in deriving a relation for the drain current (I_{DS}), let us define a quantity **V_{CB}(y), which is the channel voltage with respect to the bulk** at a distance `y' from the source end. When a drain voltage is applied and a current flows from source to drain, V_{CB} also varies continuously from source to drain. At the source end, $V_{CB} = -V_{BS}$ and at the drain end $V_{CB} = V_{DS} - V_{BS}$. The electron quasi Fermi level at the semiconductor surface (E_{Fn}) also varies continuously from the source to the drain end and is related to V_{CB} by $E_{Fn}(y) = E_F - qV_{CB}(y)$. This can be understood by considering a p-n junction at the semiconductor surface consisting of the n-type inversion layer and p-type bulk which has a reverse bias of V_{CB}. This results in E_{Fn} moving down by qV_{CB} with respect to E_F.

In the analysis of the MOS capacitor, it was argued that once strong inversion set in, the potential at the semiconductor surface should remain fixed at $\psi_S = 2\phi_B$. A similar argument holds for MOSFET too, but now ψ_S varies along the channel. $\psi_S(y)$ is calculated as the surface potential for which the electron concentration at the surface (n_{ps}) equals hole concentration in the bulk (N_A). Using eqn.(10.36), and substituting $E_{Fn}(y) = E_F - qV_{CB}(y)$ and $E_{iS}(y) = E_i - q\psi_S(y)$, it can be shown that the surface potential will be pinned at a value of $\psi_S(y) = 2\phi_B + V_{CB}(y)$, indicating a continuous variation of the surface potential. Fig. 10.24 shows the band diagram in a direction normal to surface at a distance `y' from the source end.

To compute the magnitude of the drain current, it is necessary to determine the amount of mobile electron charge in the channel. Using the relations derived for MOS capacitor and taking $\psi_S(y) = 2\phi_B + V_{CB}(y)$, we shall now calculate the mobile charge density at a distance `y' from the source end. From eqn.(10.20), the total charge density is given by

$$Q_S(y) = - C_{ox} [V_{GB} - V_{FB} - 2\phi_B - V_{CB}(y)] \text{ ……..(10.42)}$$

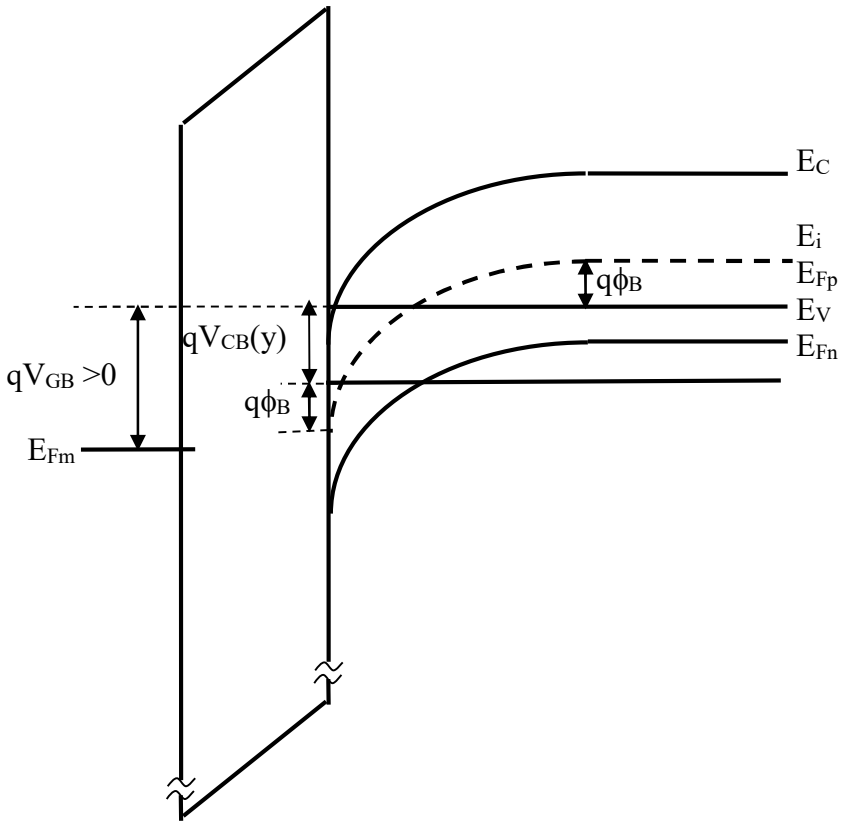


Fig. 10.24 Band diagram in a direction normal to surface at a distance y from the source

Using eqn.(10.10), the bulk charge density is given by

$$Q_B(y) = -\sqrt{2qN_A\varepsilon_s(2\phi_B + V_{CB}(y))} \text{ ……..(10.43)}$$

The electron charge density $Q_n(y)$ is now obtained as the difference of the total charge and the bulk charge. Therefore

$$Q_n(y) = Q_S(y) - Q_B(y) = -C_{ox}[V_{GB} - V_{FB} - 2\phi_B - V_{CB}(y)] + \sqrt{2qN_A\varepsilon_s(2\phi_B + V_{CB}(y))}$$

$$= -C_{ox}[V_{GB} - V_{FB} - 2\phi_B - V_{CB}(y) - \gamma\sqrt{2\phi_B + V_{CB}(y)}\,] \text{ …..(10.44)}$$

where γ is the body effect parameter given by eqn.(10.41). From eqn.(10.44) it is clear that since V_{CB} is a function of position along the channel, so is $Q_n(y)$. Let us see why this

should be so. As we move from the source to the drain, V_{CB} increases and so does the value of ψ_S required to maintain inversion. Since the gate voltage is fixed, but the semiconductor surface potential is more towards the drain end, the drop across the oxide (ψ_{ox}) obviously reduces as we move towards the drain. Since $|Q_S| = C_{ox}\psi_{ox}$, the magnitude of total charge density is less near the drain than near the source. On the other hand, since ψ_S is more towards the drain, the depletion layer width is more and hence the magnitude of the bulk charge density $|Q_B|$ is more. Now since $|Q_S|$ is less and $|Q_B|$ is more and $|Q_n| = |Q_S| - |Q_B|$, the electron concentration will be less towards the drain.

We shall now use the expression for electron charge density to derive an expression for drain current. Assuming that the current flowing in the channel under the influence of the drain voltage is purely a drift current, the electron current density at a distance 'y' from the source end is given by

$$J_n(x,y) = -qn(x,y)\mu_n \frac{dV_{CB}(y)}{dy} \quad \ldots\ldots\ldots\ldots\ldots\ldots(10.45)$$

Integrating the above equation from the surface at x = 0 to the bottom edge of the inversion layer at $x = x_{inv}$ and multiplying by the channel width `Z' gives the electron current I_n flowing from the source to the drain. Therefore

$$I_n(y) = Z\int_0^{x_{inv}} J_n(x,y)dx = -Z\mu_n \frac{dV_{CB}(y)}{dy}\int_0^{x_{inv}} qn(x,y)dx = Z\mu_n Q_n(y)\frac{dV_{CB}(y)}{dy} \quad \ldots..(10.46)$$

In the above equation, the integral of the electron concentration over the depth of the channel has been taken to be the total number of electrons per unit area in the channel. Now since there is no appreciable hole current in the channel, I_n must be constant all the way from the source to the drain end of the channel. If this were not the case, there would be rapid build up of electrons in a portion of the channel, which is not sustainable. As

$|Q_n|$ decreases from source to drain, in order to keep I_n constant, the electric field increases from source to drain and is maximum at drain end.

Since I_n is constant along the channel, we may write

$$\int_0^L I_n(y)dy = I_n \int_0^L dy = -I_D L \quad \ldots\ldots..(10.47)$$

Since the drain current (I_D) is considered to be positive when the current flows into the drain terminal, we have taken I_D= - I_n in the above equation. Now from eqn. (10.46) and (10.47), we have

$$I_D = -\frac{Z\mu_n}{L}\int_0^L Q_n \frac{dV_{CB}(y)}{dy}dy \quad \ldots\ldots\ldots (10.48)$$

Noting that V_{CB} = -V_{BS} at y = 0 and V_{CB} =V_{DS} -V_{BS} at y = L, we can now rewrite eqn.(10.48) as

$$I_D = -\frac{Z\mu_n}{L}\int_{-V_{BS}}^{V_{DS}-V_{BS}} Q_n(y)dV_{CB}(y) \quad \ldots\ldots..(10.49)$$

Now substituting eqn.(10.44) in eqn.(10.49) we have

$$I_D = \frac{Z\mu_n C_{ox}}{L}\int_{-V_{BS}}^{V_{DS}-V_{BS}} \left[V_{GB} - V_{FB} - 2\phi_B - V_{CB}(y) - \gamma\sqrt{2\phi_B + V_{CB}(y)}\right]dV_{CB}(y) \quad \ldots\ldots..(10.50)$$

The above integral yields the following expression for drain current

$$I_D = \frac{Z\mu_n C_{ox}}{L}\left[\left(V_{GS} - V_{FB} - 2\phi_B - \frac{V_{DS}}{2}\right)V_{DS} - \tfrac{2}{3}\gamma\left\{(2\phi_B + V_{DS} - V_{BS})^{3/2} - (2\phi_B - V_{BS})^{3/2}\right\}\right]$$

$\ldots\ldots..(10.51)$

This is called the **bulk charge model** and is available in the Level 2 MOSFET model in SPICE. However, this model has some drawbacks. In deriving eqn.(10.51), it was assumed that when the silicon surface is inverted, its potential is pinned at $2\phi_B + V_{CB}$. In

reality, this is the value when the surface just enters strong inversion. If the electron charge in the channel is substantial ψ_S must be greater than this value. It is true that ψ_S will never be much larger than $2\phi_B + V_{CB}$, but it may it may be more by a few times the thermal voltage (V_T). So we may say that bulk charge model consistently underestimates ψ_S. Now, if ψ_S is underestimated, $|Q_B|$ is also underestimated. On the other hand, ψ_{ox} and consequently $|Q_S|$ is overestimated. This means that $|Q_n| = |Q_S| - |Q_B|$ is definitely overestimated. Therefore, the bulk charge model consistently overestimates the electron concentration in the channel, which means that the drain current is also overestimated. Fortunately, the error involved is within 10%. A more accurate model, which considers the exact surface potential is called charge sheet model. However, this model resorts to numerical techniques and will not be discussed in this book.

Saturation region characteristics

In the above analysis, it was assumed that an inversion layer existed all the way from the source to the drain. This is true for small V_{DS}. However, as V_{DS} is increased, E_{Fn} near the drain moves lower. This means that ψ_S required to maintain inversion, should be larger. A higher ψ_S results in higher $|Q_B|$ and lower $|Q_S|$, due to lower ψ_{ox}. Therefore, with increasing V_{DS}, the electron charge density near the drain given by $|Q_n| = |Q_S| - |Q_B|$ reduces. Ultimately, at a particular value of V_{DS}, called V_{Dsat}, Q_n goes to zero. Under this condition, the channel is said to be pinched off near the drain. This might lead one to conclude that the drain current drops to zero. However, this is not true. In fact, once the electron charge at the drain end becomes very small, the electric field parallel to

the interface increases, and gradual channel approximation itself fails. Therefore, the pinch off condition at $V_{DS} = V_{Dsat}$ actually marks the failure of the gradual channel approximation rather than Q_n going to zero. However, for the purpose of modeling, V_{Dsat} is defined as the value of V_{DS} for which the gradual channel approximation gives $Q_n = 0$. Since pinch off occurs initially at the drain end of the channel when $V_{DS} = V_{Dsat}$, we can now obtain a relation for V_{Dsat} by substituting $Q_n(y) = 0$ and $V_{CB}(y) = V_{Dsat} - V_{BS}$ in eqn. (10.44). Therefore

$$V_{GB} - V_{FB} - 2\phi_B - V_{Dsat} + V_{BS} = \gamma\sqrt{2\phi_B + V_{Dsat} - V_{BS}} \quad \text{.....(10.52)}$$

Solving the above equation as a quadratic equation in V_{Dsat}, we have

$$V_{Dsat} = V_{GS} - V_{FB} - 2\phi_B + \gamma^2/2\left[1 \pm \sqrt{1 + \frac{4(V_{GB} - V_{FB})}{\gamma^2}}\right] \quad \text{.....(10.53)}$$

In order to resolve whether the plus or minus sign is to be taken in the above equation, let us consider the following argument. Since (V_{GB} - V_{FB}) equals the sum of the potential drop in the oxide and in the semiconductor, and the drop in the semiconductor at the drain end is equal to ($V_{Dsat} + 2\phi_B - V_{BS}$) when the transistor goes to saturation, we can conclude that $V_{GB} - V_{FB} > V_{Dsat} + 2\phi_B - V_{BS}$, or

$$V_{Dsat} < V_{GB} - V_{FB} - 2\phi_B + V_{BS} = V_{GS} - V_{FB} - 2\phi_B \quad \text{.....(10.54)}$$

In eqn.(10.53), the quantity within brackets is positive if we accept the plus sign and is negative if we accept the minus sign. Thus equation (10.54) can be satisfied only if we retain the negative sign. Therefore, V_{Dsat} is now given by

$$V_{Dsat} = V_{GS} - V_{FB} - 2\phi_B + \gamma^2/2\left[1 - \sqrt{1 + \frac{4(V_{GB} - V_{FB})}{\gamma^2}}\right] \quad \text{.....(10.55)}$$

As already mentioned, when $V_{DS} > V_{Dsat}$, the pinch off point moves towards the source, leaving a depletion layer between this point and the drain. The voltage (V_{DS} - V_{Dsat}) is dropped across this depletion layer, and the high electric field helps to sweep the electrons entering this region to the drain. The current in this region of the characteristics remains almost constant, a small increase resulting from the **channel length modulation**, i.e. the reduction of the effective channel length as the pinch off point moves towards the source. The modeling of the current in the saturation region will be dealt with in the next chapter.

10.3.2.2 Square Law model (Level 1 in SPICE)

Eqn.(10.51) is a relatively accurate description of current in the long channel MOSFET. However, this equation is rather cumbersome to use in hand calculations and slow to evaluate when incorporated in circuit simulation packages such as SPICE. A simplified version of eqn.(10.51) is obtained if we assume that $2\phi_B - V_{BS} >> V_{DS}$. Then we have

$$\left(2\phi_B - V_{BS} + V_{DS}\right)^{3/2} = \left(2\phi_B - V_{BS}\right)^{3/2}\left(1 + \frac{V_{DS}}{2\phi_B - V_{BS}}\right)^{3/2} \approx \left(2\phi_B - V_{BS}\right)^{3/2}\left(1 + \frac{3}{2}\frac{V_{DS}}{2\phi_B - V_{BS}}\right)$$

$$= \left(2\phi_B - V_{BS}\right)^{3/2} + \frac{3}{2}V_{DS}\left(2\phi_B - V_{BS}\right)^{1/2} \text{ .....(10.56)}$$

Substituting eqn. (10.56) in eqn. (10.51) we have

$$I_D = \frac{Z\mu_n C_{ox}}{L}\left[\left(V_{GS} - V_{FB} - 2\phi_B - \frac{V_{DS}}{2}\right)V_{DS} - \frac{2}{3}\gamma.\frac{3}{2}V_{DS}\left(2\phi_B - V_{BS}\right)^{1/2}\right]$$

$$= \frac{Z\mu_n C_{ox}}{L}\left[\left(V_{GS} - V_{FB} - 2\phi_B - \frac{V_{DS}}{2}\right) - \gamma\left(2\phi_B - V_{BS}\right)^{1/2}\right]V_{DS} \text{ .....(10.57)}$$

Substituting the relation for V_{th} given by eqn.(10.39), we now get a simple relation for the drain current given by

$$I_D = \frac{Z\mu_n C_{ox}}{L}\left[V_{GS} - V_{th} - \frac{V_{DS}}{2}\right]V_{DS} \text{ ……..(10.58)}$$

The above model is available as Level 1 in SPICE. Now, using the same approximation as above, i.e. $2\phi_B - V_{BS} >> V_{DS}$ in eqn.(10.52), we get an expression for the saturation drain voltage as

$$V_{Dsat} = V_{GB} - V_{FB} - 2\phi_B + V_{BS} - \gamma\sqrt{2\phi_B - V_{BS}} \text{ …..(10.59)}$$

Again using the expression for threshold voltage given by eqn.(10.39), we get a very simple relation for V_{Dsat} given by

$$V_{Dsat} = V_{GS} - V_{th} \text{……(10.60)}$$

Substituting the saturation drain voltage given by eqn.(9.60) in eqn.(10.58), we get an expression for the drain current when the MOSFET goes to saturation (I_{Dsat}) as

$$I_{Dsat} = \frac{Z\mu_n C_{ox}}{2L}\left[V_{GS} - V_{th}\right]^2 \text{ ……(10.61)}$$

The **square law model** gets its name from this square law type dependence of the saturation drain current on the gate voltage.

Physically, the square law model can also be derived by assuming that all along the channel the depletion width is constant, and is equal to the actual depletion width at the source end of the channel. Therefore the bulk charge density (Q_B) does not depend on the position along the channel and is given by substituting $V_{CB}(y)=V_{SB}$ in eqn.(10.43). Hence

$$Q_B = -\sqrt{2qN_A\varepsilon_{Si}(2\phi_B - V_{BS})} \text{ ……..(10.62)}$$

The electron charge density $Q_n(y)$ is now obtained following the same procedure as in the bulk charge model. Therefore

$$Q_n(y) = Q_S(y) - Q_B(y) = - C_{ox} [V_{GB} - V_{FB} - 2\phi_B - V_{CB}(y) - \gamma\sqrt{2\phi_B - V_{BS}}\ ].....(10.63)$$

Using the threshold voltage expression given by eqn.(10.39), we can simplify eqn.(10.63) to

$$Q_n(y) = - C_{ox} [V_{GB} - V_{th} - V_{CB}(y)\ ].....(10.64)$$

Now substituting eqn.(10.64) in eqn.(10.49), we get an expression for the drain current as

$$I_D = \frac{Z\mu_n C_{ox}}{L} \int_{-V_{BS}}^{V_{DS}-V_{BS}} [V_{GB} - V_{th} - V_{CB}(y)]\, dV_{CB}(y) \ .....(10.65)$$

The above integral gives

$$I_D = \frac{Z\mu_n C_{ox}}{L}\left[V_{GS} - V_{th} - \frac{V_{DS}}{2}\right]V_{DS} .....(10.66)$$

which is exactly the same as eqn.(10.58).

The square law model consistently overestimates I_D. The depletion width is more at the drain end. The assumption that the depletion width throughout the channel is the same as that at the source end results in an underestimation of $|Q_B|$. Since $|Q_n| = |Q_S| - |Q_B|$, this means that $|Q_n|$ and consequently I_D is overestimated. The overestimation is quite large and maybe as high as 20% to 30%. Only at very low drain voltages, the model is quite accurate as the assumption of a constant depletion width may be reasonably valid.

10.3.2.3 MOSFET Level 3 model in SPICE

The bulk charge model, although quite accurate, is not really suited for quick calculation of results. On the other hand, the square law model relations are quite simple,

but not very accurate. The SPICE Level 3 model was developed to fill the need for a model which is simple, and at the same time quite accurate.

To derive the Level 3 model, we introduce a variable $V_{CS}(y)$, which is the channel-to-source potential. This is related to the channel-to-bulk potential through the relation $V_{CS}(y) = V_{CB}(y) + V_{BS}$. Hence, the relation for the bulk charge density given by eqn.(10.43) can now be expressed in terms of $V_{CS}(y)$ as

$$Q_B(y) = -\sqrt{2qN_A\varepsilon_{Si}(2\phi_B + V_{CS}(y) - V_{BS})} = -\gamma C_{ox}\sqrt{2\phi_B + V_{CS}(y) - V_{BS}} \quad \ldots\ldots(10.67)$$

In this model, we take the approximation $2\phi_B - V_{BS} >> V_{CS}(y)$. Using this approximation, eqn.(10.67) can be modified as

$$Q_B(y) = -\gamma C_{ox}\sqrt{2\phi_B - V_{BS}}\cdot\sqrt{1+\frac{V_{CS}(y)}{2\phi_B - V_{BS}}} \approx -\gamma C_{ox}\left[\sqrt{2\phi_B - V_{BS}} + \frac{\frac{1}{2}V_{CS}(y)}{\sqrt{2\phi_B - V_{BS}}}\right] \quad (10.68)$$

The electron charge density $Q_n(y)$ is now obtained as the difference of the total charge and the bulk charge. Therefore from eqn.(10.42) and (10.68) we have

$$Q_n(y) = Q_S(y) - Q_B(y) = -C_{ox}[V_{GB} - V_{FB} - 2\phi_B - V_{CB}(y)] + \gamma C_{ox}\left[\sqrt{2\phi_B - V_{BS}} + \frac{\frac{1}{2}V_{CS}(y)}{\sqrt{2\phi_B - V_{BS}}}\right]$$

$$= -C_{ox}\left[V_{GB} - V_{FB} - 2\phi_B - V_{CB}(y) - \gamma\sqrt{2\phi_B - V_{BS}} - \frac{\frac{1}{2}\gamma V_{CS}(y)}{\sqrt{2\phi_B - V_{BS}}}\right] \ldots.(10.69)$$

Considering the threshold voltage expression given by eqn.(10.39), the above expression can be simplified as

$$Q_n(y) = -C_{ox}[V_{GS} - V_{th} - \alpha V_{CS}(y)] \ldots\ldots(10.70)$$

where

$$\alpha = 1 + \frac{\frac{1}{2}\gamma}{\sqrt{2\phi_B - V_{BS}}} \ldots\ldots\ldots\ldots..(10.71)$$

Now substituting eqn.(10.70) in eqn.(10.49), we get an expression for the drain current as

$$I_D = \frac{Z\mu_n C_{ox}}{L} \int_{-V_{BS}}^{V_{DS}-V_{BS}} \left[V_{GS} - V_{th} - \alpha V_{CS}(y)\right] dV_{CB}(y) \quad \text{…….(10.72)}$$

Noting that $V_{CS}(y) = V_{CB}(y) + V_{BS}$ and V_{BS} is a constant, we can rewrite eqn.(10.72) as

$$I_D = \frac{Z\mu_n C_{ox}}{L} \int_{0}^{V_{DS}} \left[V_{GS} - V_{th} - \alpha V_{CS}(y)\right] dV_{CS}(y) \quad \text{…….(10.73)}$$

The drain current in the linear region can therefore be obtained from eqn.(10.73) as

$$I_D = \frac{Z\mu_n C_{ox}}{L} \left[V_{GS} - V_{th} - \alpha \frac{V_{DS}}{2} \right] V_{DS} \text{…….(10.74)}$$

The drain voltage at which pinch off occurs at the drain end of the channel (V_{Dsat}) can be obtained as in earlier models by putting $Q_n = 0$ in eqn.(10.70). Since $V_{CS}(L) = V_{Dsat}$, we have

$$V_{Dsat} = \frac{V_{GS} - V_{th}}{\alpha} \quad \text{……..(10.75)}$$

Substituting the expression for V_{Dsat} from eqn.(10.75) in eqn. (10.74) we get the drain current at saturation as

$$I_{Dsat} = \frac{Z\mu_n C_{ox}}{2\alpha L} \left[V_{GS} - V_{th}\right]^2 \text{……(10.76)}$$

From eqn.(10.71), we see that the value of the constant α is always greater than 1. Comparing the expressions for V_{Dsat} and I_{Dsat} obtained from the Level 1 and Level 3 models in SPICE, it is quite clear that both V_{Dsat} and I_{Dsat} is lower for the Level 3 model. This is expected as the approximation $2\phi_B - V_{BS} >> V_{CS}(y)$ may be valid for a large portion of the channel and is therefore more accurate than the Level 1 model which assumes $2\phi_B - V_{BS} >> V_{DS}$. The Level 3 model is therefore a far better approximation of the Level 2 model than the Level 1 model.

Example 10.9 The MOSFET MF1 is operated with the substrate shorted to the source, i.e. V_{BS} = 0V. The channel length and width of this device is given by L = 5μm and W = 20μm. Calculate V_{Dsat} and I_{Dsat} obtained in SPICE Level 1, Level 2 and Level 3 models for V_{GS} = 5V. Assume $\mu_n = 800 cm^2/V\text{-}sec$.

For MF1, V_{th} = 0.84V, γ = 1.683 and ϕ_B = 0.35V (Ex. 10.8), $C_{ox} = 3.45 \times 10^{-8} F/cm^2$ and V_{FB} = -1.27V (Example 10.5)

Level 2 :

From eqn. (10.55), $V_{Dsat} = 5 + 1.27 - 2x0.35 + \frac{1.683^2}{2}\left[1 - \sqrt{1 + \frac{4(5+1.27)}{1.683^2}}\right] = 2.54V$

Substituting $V_D = V_{Dsat}$ = 2.54V in eqn. (10.51), we have

$$I_{Dsat} = \frac{20x10^{-4}x800x3.45x10^{-8}}{5x10^{-4}}\left[\left(5 + 1.27 - 0.7 - \frac{2.54}{2}\right)2.54 - \tfrac{2}{3}(1.683)\left\{(0.7+2.54)^{3/2} - (0.7)^{3/2}\right\}\right]$$

$$= 1.104x10^{-4}\left[4.3x2.54 - \tfrac{2}{3}(1.683)\{5.832 - 0.586\}\right] = 0.556mA$$

Level 1 :

From eqn. (10.60), V_{Dsat} = 5 – 0.84 = 4.16V

From eqn. (10.61), $I_{Dsat} = \frac{20x10^{-4}x800x3.45x10^{-8}}{2x5x10^{-4}}[5 - 0.84]^2 = 0.955mA$

Level 3 :

From eqn. (10.71), $\alpha = 1 + \frac{\frac{1}{2}(1.683)}{\sqrt{0.7}} = 2.01$

From eqn. (10.75), V_{Dsat} = (5 – 0.84)/2.01 = 2.08V

From eqn. (10.76), $I_{Dsat} = \frac{20x10^{-4}x800x3.45x10^{-8}}{2x5x10^{-4}x2.01}[5 - 0.84]^2 = 0.478mA$

10.3.2.4 Comparison of the MOSFET models

The Level 1 model uses a lot of approximations and is the simplest of all the MOSFET models. It can only be used for a rough approximation of the circuit performance. On the other hand, the Level 2 model takes much more computation time and may even result in convergence problems. The Level 3 model is a good compromise and can be used in most cases, except when highly precise results are required.

10.3.3 Process flow for a self-aligned nMOSFET

For the fabrication of an n-channel MOSFET, the starting material is a p-type silicon wafer of (100) orientation. In this example we shall discuss a self-aligned, polysilicon-gate, oxide- isolated process flow, as shown in Fig.10.25.

Step1: The first step is the isolation. Oxide isolation is standard in MOSFET technology. In this example, a recessed field oxide is used to isolate adjacent transistors resulting in better planarity. First a thin initial (pad) oxide is grown on the surface, followed by chemical vapour deposition of 0.1 μm. thick silicon nitride as shown in Fig.10.25(a).

Step2: Nitride deposition is followed by a lithography step, in which photoresist is used to protect all the areas where active transistors will be fabricated. Nitride is etched away from the remaining regions but the photoresist is not removed (Fig.10.25(b)). Using this photoresist as a mask, boron *channel stop implant* is carried out. Thus this implant is automatically aligned to the nitride free area. This prevents inadvertent and unwanted formation of a conducting channel under the field oxide. Photoresist is now removed.

Step3: Field oxidation is now carried out to realize an oxide thickness of about 1 μm. Since silicon is consumed during oxidation, only about half of this thickness (i.e. 0.5 μm)

protrudes above the surface while the rest is below the original surface as shown in Fig.10.25(c). This results in better planarity of the fabricated device. Oxidation is blocked in the regions protected by the nitride, as the oxidizing species cannot reach the silicon surface in those places. This process is therefore also referred to as *Isolation by Local Oxidation*. Silicon nitride is now removed from the active area using chemical agents like phosphoric acid. Boron threshold tailoring implant is carried out to adjust the threshold voltage of the MOSFET (this will be discussed in Chapter 11).

Step4: Gate oxidation is now carried out to realize a very thin high quality, defect-free oxide. This is the most critical step in the MOSFET processing and extreme care must be taken to ensure the high quality of the gate oxide. Polysilicon is now deposited over the entire surface by chemical vapour deposition. Using a second lithography step, polysilicon is now patterned by wet or dry etching to define the gate (Fig.10.25(d)). This step determines the channel length, one of the most important parameters in MOSFET. Dimensional control of gate lithography and poly-etching is therefore another key step in the process flow.

Step5: After the gate patterning, ion implantation for source and drain is carried out. No lithography is required for this step and hence the process is termed as *self-aligned*. The field oxide and the gate act as masks against implantation and consequently the n^+ source and drain are automatically aligned to the gate (Fig.10.25(e)).

Step6: The next step is chemical vapour deposition of phosphosilicate glass (Fig.10.25(f)). This is used as a barrier against alkali ions (which may cause instabilities in device performance). This glass is heated to around 1000^0C to flow and smoothen the contours of the device. This results in better metal step coverage.

Step7: Contact windows are now opened in the glass by the third lithography step followed by metal deposition and patterning (Fig.10.25(g)).

Step8: Finally a passivation layer (usually silicon nitride) is deposited on the finished device. A final lithography step is then carried out and only the bondpads are exposed.

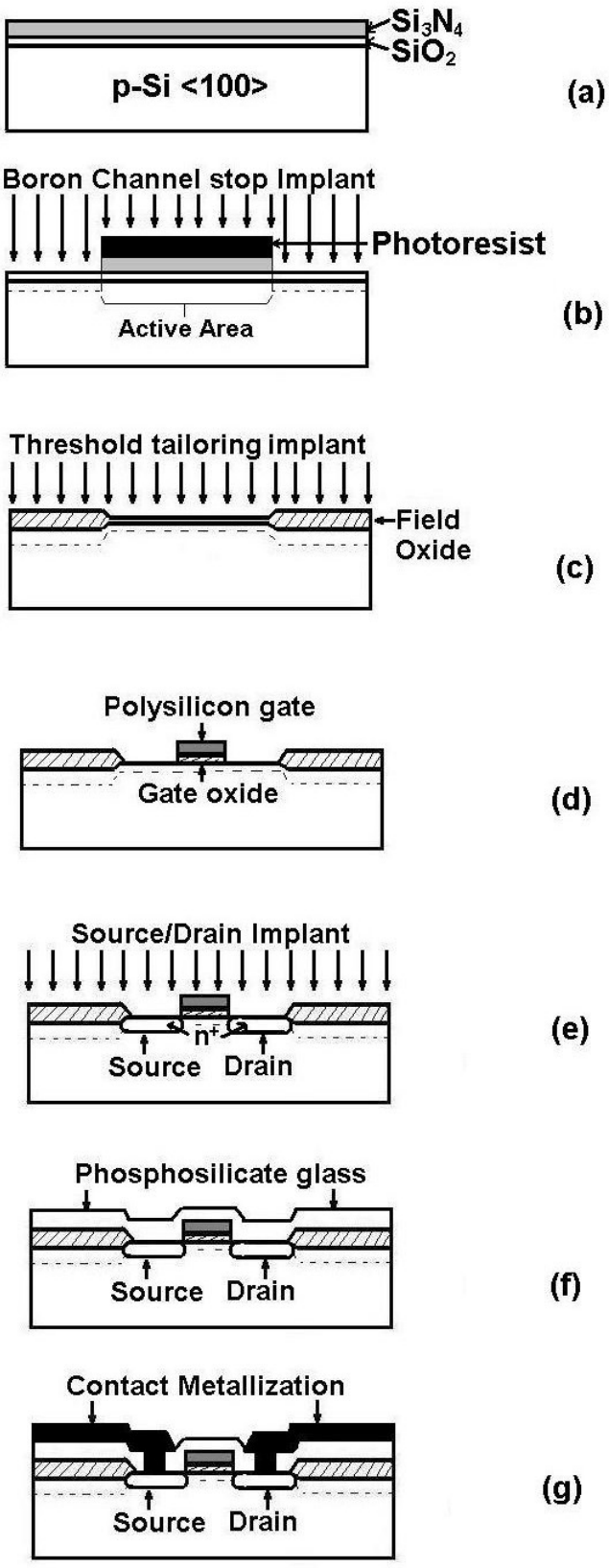


Fig.10.25 Process steps for fabricating a self-aligned MOSFET for Integrated Circuit

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PROBLEMS AND QUESTIONS

1. A MOS capacitor is fabricated on p-silicon substrate with a doping concentration of 10^{16} cm^{-3}. If the gate oxide thickness is 500 Å, and n^+ polysilicon is deposited as gate metal, find
   (a) The maximum capacitance/area
   (b) The minimum capacitance/area
   (c) The ideal threshold voltage assuming $V_{FB} = 0V$
   (d) The threshold voltage taking the work function into account.
   (e) If the experimental C-V characteristics of this capacitor show a V_{FB} of –1V, what is the amount of fixed oxide charge?
2. If donor-like interface states are present, how will they affect the high frequency C-V plot of a MOS capacitor on n-Si substrate? Explain with sketches.
3. (a) An n^+ polysilicon gate MOS capacitor was to be fabricated so that it had a threshold voltage (V_{th}) of 1V at 300K. The available silicon wafers were uniformly doped with a concentration of $N_A = 2 \times 10^{16}/cm^3$. It was assumed that the oxide and interface charges would be so small that they would not affect V_{th}. What is the required gate oxide thickness?

   (b) After fabrication of the MOS capacitor, V_{th} was measured to be 0.65V. The capacitor was then subjected to bias heat stress. First, it was heated to 250°C with +30V applied to the gate, so that all the mobile ions (which are positively charged) moved to the Si-SiO_2 interface. After cooling, V_{th} was measured to be 0.5V. Again, the capacitor was heated to 250°C, this time with -30V applied to the gate, so that the ions are attracted to the polysilicon- SiO_2 interface. Now, after cooling, V_{th} was

measured to be 0.7V. Estimate the values of Q_f/q and Q_m/q. Where were all the mobile ions located initially, assuming they were all concentrated in a plane parallel to the interfaces. Assume Q_{it} and Q_{ot} to be negligible.

4. (a) A MOS diode is fabricated on a p-type silicon substrate having a doping concentration of $N_A=5x10^{15}/cm^3$. The gate oxide thickness (t_{ox}) is 200Å. If n^+ polysilicon is used as the gate material for this device and it has a fixed oxide charge density given by $Q_f/q = 1.0x10^{11}/cm^2$, calculate

   (i) the maximum depletion width (W_{max})

   (ii) the flat-band voltage (V_{FB})and

   (iii) the threshold voltage (V_{th}) for this device.

   (b) A MOSFET is fabricated on the same substrate with the same gate oxide thickness and fixed oxide charge as in (a). This MOSFET also has n^+ polysilicon gate and is operated with V_{BS} = -2V. The channel length and width of this device is given by L = 1μm and W = 5μm. Calculate V_{Dsat} and I_{Dsat} using SPICE Level 3 model for V_{GS} = 5V.

5. Matched p and n-channel MOSFETs are to be fabricated in the same silicon wafer (i.e $V_{thp} = -V_{thn}$ for V_{BS} =0). To do this, an n-Silicon sample with doping concentration $N_D = 10^{15}/cm^3$ is taken and a p-well is diffused in it. Identical p and n-channel MOSFETs are now fabricated in the n and p-regions respectively of this wafer. The gate oxide thickness is 1000Å, $Q_f/q = 3x10^{11}/cm^2$ and Aluminium gate is used. What is the doping concentration of the p-well to obtain matched devices? What are the values of V_{th} for the FETs, if V_{BS} = 0?

6. An n-channel MOSFET has the following specifications : channel width (Z) = 5μm, channel length (L) = 1μm, gate oxide thickness (t_{ox}) = 500 Å , electron mobility = 800 cm^2/V.sec, threshold voltage (V_{th}) = 1V.

   Find the drain current flowing through the device for the following conditions:

   (i) V_{GS} = 2V, V_{DS} = 0.5 Volts

   (ii) V_{GS} = 2V = V_{DS} Using (a) SPICE level 1 and (b) level 3 model

7. (a) A n^+ poly-gate n-channel MOSFET has a substrate doping concentration of $10^{16}/cm^3$ and a gate oxide thickness of 1000Å. What will be the change in its threshold voltage if the source-to substrate voltage (V_{SB}) is increased to 2V from zero?

   (b) The above devices are isolated by field oxide in an integrated circuit. For safe operation upto 10V, what should be the minimum thickness of the field oxide so that the FETs remain isolated?